

Weeks 2-10

Facilitator guides

Preparation, pacing, hints, proofs, source notes, and extra-page solutions.

Grade bands are entry points. Begin with objects, then try to explain what happens.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Keep facilitator solutions separate from student pages.

Week and investigation	PDF pages
02 Switches and lamps	2-6
03 Repeating secret machines	7-11
04 Around the ring	12-16
05 Tower cities	17-21
06 Code-breaking with guarantees	22-27
07 Take-away games	28-32
08 Rooks and piles	33-38
09 Bouncing paths	39-44
10 Bridges and delivery routes	45-50

Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

Weekly footers retain their own page numbers and activity IDs for the use log.

Revised September 27, 2026. Unpiloted revisions; record actual use after teaching.

Prepare the toggle lab

v3 revision, September 27, 2026: unpiloted. Week 1 feedback motivates more contrasting attempts, clear recording, and later introduction of representations. No Week 2 classroom outcomes have been reported.

Materials and readiness

For seven children, bring 29 two-sided counters plus 8 spares, pencils, and scratch paper; print the first page for three K, three middle, and one upper child (seven starting sheets). Keep continuation masters ready and copy selected pages as needed; the extra can stay digital until chosen. Paper disks with one shaded face work. Keep counters on lamps and flip their faces.

K: no reading; distinguish faces and pair/count to four. **Middle:** labels 1–4, count to four; pairing can replace odd/even vocabulary. **Upper:** count to six and track yes/no decisions. **Extra:** follow paths, count targets within a piece, and reason about a cut. Adult scribing removes handwriting barriers.

Handle, then launch: 0–15 minutes

Let everyone handle counters and make pictures freely during 0–10 minutes. At 10–15 minutes, gather around one four-lamp mat. Show all OFF, press 12, then press 23 so children see an ON lamp flip OFF. Have a child replay it. Point to the two ON lamps and write the move list “12,23.” Reset and replay the record. State the goal: make a chosen picture by flipping the two ends of a line. Distribute different starting tasks only after this shared legal attempt. Do not announce parity yet.

A flexible hour

15–35: K Problems 1–4; middle 1–3; upper 1–2. Adult A anchors K; organizer checks that older children can replay a list. Compare a successful and an unresolved target when useful. **35–40:** stand, stretch, reset. **40–55:** continue the same idea or choose the next stage below. **55–60:** share one attempt and tidy. These are proposed intervals; do not rush to finish pages.

Stops and transitions

K: a satisfying stop is making and undoing adjacent, opposite, and four-lamp targets. Offer odd targets only after the move is reliable. Give several attempts before discussing pairing; the new lamp-1 button comes last.

Middle: stop after contrasting target experiments. Introduce the two-end table (4) to examine a difficulty they encountered. Demonstrate a four-circle record before collection (7); the collection and completeness proof can wait.

Upper: stop after finding shorter lists for the three targets. Compare cancellation and complementary sets before introducing forced choices (6–8). The six-ring is a separate comparison. **Extra:** allow a stop after component examples; join components before tree cuts.

Prompts for adult conversation

“What changed at each end?” “Can your partner replay it?” “What did a middle lamp do twice?” Ask for impossibility, completeness, and shortestness explanations when welcome. Pages 2–5 give proofs and added checks; pointing or moving counters can carry the argument.

Parity and all the pictures

K–1 solutions and explanations

Two identical presses undo each other; replaying any list in reverse undoes it. Top/right (1,2) needs 12. Top/bottom (1,3) uses 12 then 23. A single ON lamp is impossible under the pair rule. Both OFF endpoints become ON (+2), both ON become OFF (-2), or one ON trades places (0). Starting at zero, the number ON stays even. Children may demonstrate these three cases and pair the counters instead of saying “parity.” The new one-lamp button breaks this invariant and makes the singleton immediately.

Middle: the complete set

The eight states, listed by their ON lamps, are:

Number ON	Possible sets
0	none
2	12, 13, 14, 23, 24, 34
4	1234

Here “13” in the *state list* means lamps 1 and 3, not a legal edge. Avoid conflating a target list with a move list. All six pairs can be made by a path on the square. Each interior vertex is touched twice and returns OFF; each end is touched once. To make all four, use edges 12 and 34. No odd-size state is possible. This proves both that the list is complete and that every listed state is reachable. There are 12 blank recording boxes so their count does not give away the answer.

Why paths generalize

Pair the desired ON vertices. For each pair, choose a path and press its edges. At an intermediate vertex, contributions cancel in pairs; endpoints give one flip. Shared edges or vertices cause no problem: all flip counts add modulo two. An edge pressed twice can be deleted from the list. On any connected graph, exactly the even targets are reachable. Connectivity matters; see the extra.

What is being added?

Old shape addition used union: a shared stroke was kept once. Toggling is **symmetric difference**: something present twice cancels. This new operation restores information in the identity $(A \triangle B) \triangle B = A$. A one-minute stick-picture comparison is an optional bridge for returners, not a second core activity. Lowell Handout 11, problem 11.3, asked for splitting and recombining shapes; Handout 9, problem 9.1, used door toggles indexed by divisors. Here the adjacent-pair move and reachability problem are new.

Two answers, then an optimum

Upper Problems 1–2: constructions plus lower bounds

Target $\{1,3\}$: edges $\{12,23\}$ or $\{51,45,34\}$. Minimum **2**. One press can light only adjacent lamps; these are not adjacent. Target $\{1,2,3,4\}$: $\{12,34\}$ or $\{23,45,51\}$. Minimum **2**, because one press changes at most two lamps. Children should supply both the successful move list and the reason one press cannot suffice.

Upper Problems 3–8: exactly two reduced solutions

For target $\{1,3\}$, the decision table is:

	51	12	23	34	45	lamp 5
First no	no	yes	yes	no	no	OFF
First yes	yes	no	no	yes	yes	OFF

Once 51 is chosen, lamp 1 forces 12, lamp 2 forces 23, and so on. There can be at most two solutions using each edge zero or one times. Complementing an edge set preserves the target, because each lamp has exactly two incident edges: a flip count of 0, 1, 2 becomes 2, 1, 0, with the same parity. Thus every reachable target has exactly two reduced solutions. There are infinitely many unreduced move lists if repeated presses are permitted; state the distinction. For target $\{1\}$, first-no forces only 12,23,34,45, leaving lamp 5 ON incorrectly; first-yes forces only 51, again leaving lamp 5 ON. Both branches fail. Parity explains the same obstruction globally.

Upper Problems 9–10: the five-ring speed limit

An even target is reachable by paths. Its two reduced press sets have sizes k and $5 - k$; at least one is at most 2. Any repeated presses can be cancelled without changing the target, so the smaller reduced set is globally shortest. On an odd ring the two sizes never tie. On the six-ring, opposite lamps 1 and 4 have solutions $\{12,23,34\}$ and $\{61,56,45\}$, each size 3. The same forced-choice proof shows there is no third reduced set, hence no shorter route. More generally the two cycle solutions have sizes $k, n - k$.

Undergraduate interpretation

Write OFF=0 and ON=1. Each edge is a column with two 1's in a vertex-edge incidence matrix. A move set solves $Ax = t$ over the two-element field. The all-edge cycle is a nonzero null vector; on a single cycle it is the only one. Our forced-choice proof is binary elimination without matrix notation. The connected-graph image is the even-parity subspace; its dimension is $n - 1$.

Networks and real optimization

Extra: every component matters

Target $\{A,D\}$ is impossible: the triangle and the separate edge each contain an odd number of target vertices, while every move changes two vertices in one component. Target $\{A,C,D,E\}$ uses AC and DE, minimally two presses. In general, a target is reachable *if and only if each component contains an even number of targets*. Necessity is component parity; sufficiency pairs targets within each component and uses paths. Isolated vertices are components too.

Extra: why a tree has one solution

Delete an edge e . Count desired ON vertices on either side. Presses wholly on one side affect its parity by zero; the only move crossing the cut is e . Thus e is pressed an odd number of times exactly when that side has an odd target count; otherwise it is pressed an even number of times (possibly zero). The two sides agree because the whole target is even. These decisions uniquely specify every edge. Existence follows from the path construction. They therefore give the unique reduced solution and a shortest move list; repeated edges can only lengthen it. A leaf-removal algorithm gives another constructive explanation.

Research connection and its boundary

The pressed-edge set has odd degree exactly at the target vertices T : this is a **T-join**. The children's optimization is exactly minimum-cardinality T-join on a cycle. General networks can have many cycle choices; edge costs lead to the minimum-weight T-join problem and route-inspection algorithms. We prove the elementary parity and tree/cycle cases here, not the general matching algorithm or its polyhedral theorem. **Edmonds and Johnson (1973)**, *Matching, Euler tours and the Chinese postman*, *Mathematical Programming* 5, 88–124. The publisher abstract identifies matching, Euler tours, and the optimization result: doi:10.1007/BF01580113.

Cornuéjols, *Combinatorial Optimization: Packing and Covering*, Chapter 2, pp. 27–29, defines T-joins, explains shortest-path/matching reduction, and states the Edmonds–Johnson theorem: author's monograph. It uses acyclic representatives; a cycle can be deleted without changing odd-degree vertices and without increasing nonnegative cost.

Undergraduate anchor: MIT 18.06SC, Graphs, Networks, Incidence Matrices. Its usual real-valued matrix framework motivates the representation; the binary adaptation and proof above are ours.

Teaching sources and use history

Math Circle by the Bay, preface pp. viii–x, advocates themes with deep mathematical context, manipulatives, varied pace, and reserve challenges. Rozhkovskaya, Lesson 3 “At the lesson,” item 1, lets children attempt and explain before a table is introduced; Lesson 7, item 1, warns how missed moves can distort a game investigation. Our staffing and timing are adaptations. Record actual tasks as F02-K/M/U/X-v3; preparation is not evidence of teaching. Reserve the general graph version for a later year if unused. **Verification:** `verify.py` enumerates move subsets for 4-, 5-, and 6-cycles, checks two complementary solutions per even target, minima, and the extra examples. The arguments above stand independently of computation.

v3 checks and observations

Added student instances

K 3: targets 14,24,1234,empty use 14; 12,14; 12,34; no moves. **K 5:** the three targets have odd size and all fail under pair flips. **K 6:** the new button flips only lamp 1. It makes target 1 immediately; target 123 uses that button and edge 23; target 2 uses the button and 12. Thus children can test all three contrasts without changing the button's location.

Middle 2–3: targets 12,13,24,1234 use 12; 12,23; 12,14; 12,34. Target 23 uses 23; target 14 uses 14. Single 1 and triple 123 fail. **4:** endpoint counts are 0 to 2, 2 to 0, and 1 to 1. **6:** path 12,23,34 leaves 1,4 ON; path 14,43 leaves 1,3 ON. Edge 43 is the same as 34. **7–8:** the complete eight states and path proof are on page 2. Ask for completeness orally, after the collection is useful.

Upper 2: adjacent target 12 has lists 12 and 23,34,45,51; its minimum is 1. The other two minima are 2 (page 3). **3:** all three lists yield target 13. **4:** complementary pairs have counts 1/4,2/3,2/3,3/2, with targets 12,13,1234,14 respectively. **6–8:** forced rows for target 1234 are (51,12,23,34,45)=(no,yes,no,yes,no) and its complement; singleton failure is on page 3. **10:** six-ring target 14 has two lengths 3/3; target 13 has 2/4. Use this to distinguish ties from the odd-ring case.

Extra: new construction checks

1: AC; DE; impossible; AC,DE; impossible; AB,DE. The failures AD and ABCD have odd target count in each component. **3:** with bridge CD, AD uses AC,CD; ABCD uses AB,CD. **4:** paths give AC,CD,BC,CD; cancel both CD presses to leave AC,BC, hence AB as desired. **6:** tree targets AC and ACEF use AB,BC and AB,BC,DE,DF. **7:** named-side target counts are 1,1,2,1,1, so BD is unused and the other four are used. Proof of every tree target is on page 4.

Record evidence before revising again

Record date, actual problems/targets attempted, successful replay of the rule, which records needed adult demonstration, children's explanations, and what they wanted to continue. Separate observed actions or quotations from interpretations and next-time proposals. Mark each result tried, conjectured, checked in cases, proved, or supplied. Record v3 IDs only if used. The page expansions and gates are design hypotheses, not classroom findings.

Prepare the code machines

v3 revision, September 27, 2026: unpiloted. Week 1's reported need for more concrete work and clearer sequencing is the evidence for this redesign; it is not evidence that these code tasks have been tested.

Materials and prerequisites

Give K children two sets of circle/triangle/square slips, middle children A–D cards, and upper children A–E (F–I in reserve). Bring 13 counters, paper and pencils. Print first pages for three K, three middle, and one upper child (seven starting sheets). Keep continuation masters ready; copy later pages when chosen. The extra can remain digital. Read aloud and scribe freely.

K: recognize shapes and count to three. **Middle:** recognize letters as labels, count to six, and follow a fixed mapping. **Upper:** track five positions and coordinate return times; multiples can be generated by moving counters. **Extra:** common multiples to 20 and organizing splits by largest part. Factorials and formal group notation are not prerequisites.

Handle, then launch: 0–15 minutes

Let children arrange cards into messages during 0–10 minutes. At 10–15 minutes, gather around the circle-to-triangle-to-square-to-circle key. Point to one input, follow its arrow, and place its output below it. Encode a three-symbol message one position at a time, keeping the original row until the new row is complete. Have a child read the output and check it against the arrows. Explain that the next turn uses the new row as its input. Then distribute starting tasks. Do not introduce loop classification yet.

A flexible hour

15–35: K sends and reverses messages; middle experiments with messages then P/Q; upper compares five-loop and 3+2 keys. Adult A stays with K; organizer checks complete-row substitution. Compare a first return when useful. **35–40:** stand, stretch, reset. **40–55:** repeat, draw loops, or choose an extension when ready. **55–60:** share a key and its return, then tidy. Give one page at a time.

Stops and representation gates

K: sending and decoding several messages is a complete stopping point. After repeated single-symbol turns, introduce the arrow loop as a record of those moves. The swap/fixed key comes later; the broken key is optional.

Middle: stop after P/Q and a four-turn construction. Demonstrate how one row of the key becomes an arrow before drawing all loops. Let children build each size arrangement before asking whether the cases are complete.

Upper: stop after the 5 versus 3+2 comparison. Move to simultaneous-return markings, then classification (6–7). Composition (8–11) is a separate direction. **Extra:** stop after constructing the 15-turn key; classification is the route to its upper bound, nine letters a later reserve.

Proof conversations

Ask “Is every letter home?” “Can two inputs point to one output?” “What can be left after the largest loop?” Keep the loop theorem, exhaustive bounds, and commuting explanations in conversation; the full arguments follow. Do not read a supplied result as if the child proved it.

Reversibility and loops

K-1 checked results

The three-shape key sends circle–triangle–circle to triangle–square–triangle. The input giving square is triangle. Every symbol returns after three turns; the order of names or the starting point does not affect that count. Under the changed key, circle and triangle swap while square stays put. All are home after two turns; square is also home after one. A key sending both circle and triangle to square loses information: the receiver cannot determine which was sent.

Middle Problems 1–5

P sends ABBA to **BCCB**; decoding CAAC gives **BCCB**. A one-to-one key preserves equality and inequality between positions, so ABBA cannot become ABCD. Its first and last outputs must match, as must the middle pair.

Turn	P	Q
0	ABCD	ABCD
1	BCAD	BADC
2	CABD	ABCD
3	ABCD	BADC
4	BCAD	ABCD

P has a three-cycle A–B–C–A and fixed D, so order 3. Q has two swaps A–B and C–D, so order 2. A single fixed letter does not certify that the whole key has returned.

Middle Problems 6–8: complete classification

A four-cycle A–B–C–D–A has order 4. The possible cycle-size partitions and their orders are:

Loop sizes	4	3+1	2+2	2+1+1	1+1+1+1
Order	4	3	2	2	1

These exhaust four letters by sorting on the largest loop. Order 6 is impossible with four letters; separate loops of size 3 and 2 require five. Counting all possible keys gives $4 \cdot 3 \cdot 2 \cdot 1 = 24$, but the classification uses five cases. Do not replace the investigation with a factorial exercise.

Why every reversible key splits into loops

Follow arrows from any symbol. Finiteness forces a repeated symbol. The first repeat must be the starting symbol: if a later point were first repeated, it would have two distinct predecessors, violating the one-to-one rule. Remove that loop and repeat with an unused symbol. Loops are disjoint and exhaust all symbols. On a loop of length a , the first return is a and all returns occur at multiples of a . All loops return together at their least common multiple. This is a child-accessible proof, not just a pattern from examples.

A maximum and noncommuting keys

Upper: the five-letter example

Successive messages are ABCDE, BCAED, CABDE, ABCED, BCAED, CABED, ABCDE. The 3-cycle returns on turns 3,6,9,...; the 2-cycle on 2,4,6,... The first common return is **6**. A five-cycle takes only 5, so the longest individual loop need not be optimal.

Sizes	5	4+1	3+2	3+1+1	2+2+1	2+1+1+1	1+1+1+1+1
Order	5	4	6	3	2	2	1

Let children organize the blank table before showing the complete list below. Largest loop 5 or 4 leaves just one split; largest 3 leaves 2 or 1+1; largest 2 leaves 2+1 or 1+1+1; largest 1 forces five singletons. Ask why these exhaust the leftovers. This complete list plus the loop theorem proves no five-letter key exceeds 6; the displayed key attains it. The mathematical work is construction *and* upper bound. If reading the full table is a barrier, arrange the five letter cards into loops and enumerate the same seven cases physically.

Upper: P then Q is not Q then P

P swaps A/B; Q swaps B/C; both fix D/E. P then Q takes ABCDE to BACDE to **CABDE**. Q then P takes ABCDE to ACBDE to **BCADE**. At letter A alone: A goes to B then C in the first order, but A then B in the second. Thus the maps differ. The composite P then Q is the loop A–C–B–A with D/E fixed: its order is **3**, even though each factor has order 2. To undo it, apply Q then P. Disjoint swaps, such as A/B and D/E, commute: neither changes any input moved by the other. Their common composite has two 2-cycles and order 2.

Extra: the eight-letter optimum is 15

Use a 5-cycle on A–E and a 3-cycle on F–H. Their first common return is 15. Exhaustive upper bound by largest loop: 8 gives 8; 7 leaves one and gives 7; 6 leaves at most a 2-cycle and gives 6; 5 leaves three letters, giving at most $\text{lcm}(5, 3) = 15$; 4 leaves four, with largest possible LCM 12 from 4+3+1; if all loops have size at most 3 their LCM divides 6. Thus nothing exceeds 15. For nine letters, 5+4 gives **20**. For completeness: largest 9,8,7,6 give maxima 9,8,14,6; largest 5 leaves at most four and gives at most 20; largest 4 gives at most 12; all at most 3 gives at most 6. Adding a letter never decreases the maximum (fix the new letter) but need not strictly increase it: sizes 5 and 6 both have maximum 6. On six letters, partitions with largest 6,5,4,3,2,1 have maximal orders 6,5,4,6,2,1.

The exact research question

Undergraduate mathematics

A reversible substitution on n symbols is a permutation, an element of the symmetric group. A repeated key is a power of this permutation; its return time is its order. Disjoint cycles, least common multiples, inverses, and noncommuting products are standard abstract algebra. No formal group axioms are needed in the child-facing activity. The words “key” and “code” describe the entry point; these small deterministic substitutions are not modern secure encryption. **Thomas Judson**, *Abstract Algebra: Theory and Applications*, Chapter 5, §5.1 “Definitions and Notation,” especially disjoint-cycle examples and Theorem 5.9 on disjoint-cycle decomposition. See university-hosted textbook, printed pp. 59–63. Our explicit loop proof supports the activity independently of theorem numbering.

A direct research lineage

Marc Deléglise, Jean-Louis Nicolas, and Paul Zimmermann, *Landau’s function for one million billions* (2008 arXiv manuscript), §1, defines $g(n)$, the maximum order of a permutation of n symbols, and relates it to least common multiples of partitions. The paper develops algorithms for very large n : arXiv:0803.2160; paper, introduction pp. 1–3. The classroom’s $g(5)=6$ and $g(8)=15$ are *exact small instances of the same optimization problem*. We do not claim that finding these small values is new research. Efficient computation and large- n growth are the research directions. The extra packet asks for a proved maximum, not a decorated encoding exercise.

Downloaded sources and pedagogical choices

Rozhkovskaya, *Math Circles for Elementary School Students*, Lesson 2, problems 2.2–2.8 (EPUB section 12), uses secret codes and shifts. We retain the physical key and short-message entry while replacing the core with arbitrary reversible substitutions and loop structure. This week’s keys need not be shifts; week 4 imposes a fixed circular order. *Math Circle by the Bay*, preface pp. viii–x, supports a deep theme accessible across levels, manipulatives, explanation, and spare challenges. Rozhkovskaya Lesson 3, “At the lesson” item 1, supports attempted solutions and explanations before tabular organization. Our exact keys, tables, staffing, and timing are new adaptations.

Prior use and verification

Lowell Handout 8, problem 8.1, already used divisibility for distributing apples among 3,4,5,6 guests. The LCM idea here is revisited through synchronized permutation loops, with new questions about maximal orders. No saved use log proves these particular machines were taught. Record actual instances as F03-K/M/U/X-v3 and note which keys or partitions children saw. `verify.py` checks every permutation for $n=4,5,6,8,9$ against the claimed maxima and verifies all message traces and compositions. It is a separate check of the finite results, not a substitute for the loop and partition proofs.

v3 checks and observations

Added concrete instances

K 2: circle–triangle–circle becomes triangle–square–triangle; square–square–triangle becomes circle–circle–square; triangle–circle–square becomes square–triangle–circle. **3:** received triangle–circle–triangle decodes circle–square–circle. **4–6:** each start returns at turn 3; the whole row just before return is square–circle–triangle. **7–8:** swapped shapes return in 2, the fixed square at every turn; whole return is 2 versus 3 for the original key. **9:** either circle or triangle could have produced square.

Middle 1: P gives BCCB, BDBD, BCAD. **2:** CAAC decodes BCCB; DBDB decodes DADA. **3:** every key preserves the equal-position pattern of ABBA. A child may test many keys; the proof is that identical inputs share an output and distinct inputs have distinct outputs. **4–5:** P/Q rows and loops are on page 2. **6–8:** four-letter loop returns at 4; 3+1 at 3; 2+2 and 2+1+1 at 2; four fixed letters at 1. These five partitions rule out 6; five letters with loops 3+2 achieve 6.

Upper: experiments before classification

1: R rows are ABCDE, BCDEA, CDEAB, DEABC, EABCD, ABCDE; S rows are on page 3. **3:** 4+1 returns at 4; 2+2+1 at 2; 3+1+1 at 3. **4:** size 3 marks 3,6,9,12; size 2 marks 2,4,6,8,10,12. First common return is 6. Introduce “least common multiple” as a name for this record, not an entry requirement. **6–7:** the seven partitions and completeness proof are on page 3. Do not disclose the count before children organize cases. **8–11:** page 3 gives both compositions and inverse; P/T both orders give BACED.

Extra: bounds after constructions

1–2: returns for 8,4+4,6+2,5+3 are 8,4,6,15. Two loops of equal length return together at that length, not its square. **3–4:** for largest loops 8,7,6,5,4, at most 3, row bounds are 8,7,6,15,12,6. In the last row, every loop has size 1,2,3, so turn 6 returns all loops; no list of ten partitions is necessary. **5:** nine-letter examples 5+4,7+2,4+3+2 return at 20,14,12. **6:** the nine-letter bounds are on page 3. **7:** six-letter trials return 6,6,5. The six-letter partition bound on page 3 proves the plateau at $g(5)=g(6)=6$; examples alone do not prove a maximum.

Observe, then adapt

Record actual keys/problems used, whether every letter was changed from the old row, a child’s return prediction, success reading/drawing loops, copying that displaced exploration, and requests to continue. Keep observations separate from explanations of why they happened and next-time proposals. Mark proof status explicitly. All v3 expansions are unpiloted design proposals.

Prepare the ring shifts

v3 revision, September 27, 2026: unpiloted. This applies Week 1's classroom guidance to new ring and shuffle work; no outcomes from teaching this revision are known.

Materials and prerequisites

Print first pages for three K, three middle, and one upper child (seven starting sheets). Keep continuation masters ready to copy as needed, including the later ring mats. Leave the extra in reserve. Bring seven counters, pencils, scratch paper, and cards 0–7; cards 8–15 extend the reserve. A paper strip numbered 0–7 distinguishes card places from labels. No wheels or brads are required.

K: match four shapes, count hops to three with support. **Middle:** labels A–J, count to ten; read-aloud works. **Upper:** count to eighteen and compare routes; general gcd proof additionally requires division and adult-supported coprime divisibility. **Extra:** track positions and repeated doubling with remainders; exponent notation is optional.

Handle, then launch: 0–15 minutes

Give 0–10 minutes to explore counters on rings. At 10–15 minutes, gather around the four-shape mat. Define the arrow direction by moving circle to triangle. Show a two-hop turn, counting only movements, and point to square as the landing; triangle was passed over. Reset and let a child replay it. Draw only the landing after the starting circle. Take one more two-hop turn and record its landing. Now distribute different starting tasks. The purpose is a shared action and record, not a gcd explanation.

A flexible hour

15–35: K compares starts under one/two hops; middle uses the common shift and records two routes; upper explores +4 and +8. Adult A anchors K; organizer checks landing records with older children. Compare routes when useful. **35–40:** stand, stretch, reset. **40–55:** continue experiments or introduce one new record when useful. **55–60:** share and tidy. Page counts are reserves, not a completion schedule.

Stops and later directions

K: stop after one-hop and two-hop comparisons; three hops and hidden-turn undo are later choices. **Middle:** stop after +2 routes from A,B,C,D. Marking the two sets helps explain a split already encountered; +3,+5 and inverse checks follow. The optional complete list of full-visit steps is an adult extension.

Upper: +4/+8 loops are a satisfying stop. Compare +3,+6,+5 before the summary table (7). Skip filling known entries if copying slows the child. Offer distance records only after physical routes; the strip/gcd proof is optional adult work. **Extra:** obtain several real rows and position tracks before the map, loops, or doubling rule. Stop at the eight-card return if that is enough.

What to listen for

“Did you land there or pass over it?” “Which start is unused?” “Is this a card label or its place?” All shifts can be undone; a short orbit does not mean an undecodable key. Discuss why an argument covers every start only after children can show examples. Proofs and checked outputs follow.

Exact routes and the gcd rule

K-1

From circle, one hop visits triangle, square, star, circle (four turns); two hops alternate circle/square (two turns), never landing on triangle or star. A secret circle-to-square shift is two hops, so triangle becomes star, square becomes circle, and star becomes triangle. Two hops undo themselves. Three-hop turns follow circle, star, square, triangle, circle, visiting all four. On six positions, among steps 1 through 5, only 1 and 5 visit all six; 2 and 4 give two loops of length 3; 3 gives three loops of length 2.

Middle

A to D is +3 on A-J. GDG decodes to **DAD**; BAG encodes to **EDJ**. B becomes E, so the additional B-to-F clue contradicts the fixed-shift rule. Repeated +2 gives A,C,E,G,I,A and B,D,F,H,J,B. Marking alternate letters explains why the two routes cannot meet: adding two preserves that marking. Repeated +3 gives A,D,G,J,C,F,I,B,E,H,A. The steps visiting all ten are **1,3,7,9**. Any shift is reversible: +2 is undone by -2, or +8. Skipping some letters along one repeated orbit is not information loss.

Upper: key and complete classification

9 to 1 gives +4, which takes 2 to 6. Outputs 1,4,10 decode to **9,0,6**. Its four loops are (0,4,8), (1,5,9), (2,6,10), (3,7,11), each returning in three turns.

Shift k	0	1	2	3	4	5	6	7	8	9	10	11
Return time	1	12	6	4	3	12	2	12	3	4	6	12
Number of loops	12	1	2	3	4	1	6	1	4	3	2	1

Steps 1,5,7,11 visit all twelve. The zero rule is the identity and has first *positive* return time 1, not zero. For eighteen positions, +6 returns after 3; +5 after 18.

Why the formula works

For +8 on 12 places, return means $3 \mid 2t$. Draw three t-strips, totaling $3t$. If the first two strips total a multiple of 3, removing them leaves t a multiple of 3. Thus the first positive return is 3. This proves this case for every t . The optional general bridge is on page 4; do not treat a completed table as its proof.

From turning to shuffling

Upper: inverse and composition

On twelve places, +4 is undone by +8 (or -4). For every starting x , +4 followed by +7 gives $x+11$ modulo 12; reversing the order gives the same because integer addition commutes. Every shift is bijective, even if its orbit from 0 is a proper subset. The visit-everywhere condition is $\gcd(n, k) = 1$, not a condition for reversibility. This distinction is central: a substitution permutation can have several cycles.

Extra: eight cards, three shuffles

Rows, left to right with the leftmost card the top:

Shuffle	Row
0	0 1 2 3 4 5 6 7
1	0 4 1 5 2 6 3 7
2	0 2 4 6 1 3 5 7
3	0 1 2 3 4 5 6 7

The old-to-new position map is **0,2,4,6,1,3,5,7**. Its cycles are fixed 0, fixed 7, (1,2,4), and (3,6,5). Thus the whole shuffle returns after 3 and not earlier. This is also a direct week 3 loop argument. For old position x in the first half (0–3), the new position is $2x$. In the second half (4–7), it is $2(x - 4) + 1 = 2x - 7$. For x below 7 these are exactly $2x \bmod 7$. Position 7 stays 7, separately; it must not be merged with residue 0. Repetition gives $2^t x \bmod 7$. The powers 2,4,8 have remainders 2,4,1, so the first identity occurs at $t=3$. Tracking position 1 rules out earlier global returns, and remainder 1 fixes all positions 0–6.

Extra predictions

Six cards use modulus 5. Powers 2,4,8,16 give remainders 2,4,3,1, so the order is **4**. Sixteen cards use modulus 15: 2,4,8,16 give 2,4,8,1, again order **4**. In general an out-shuffle on $2m$ cards maps positions below $2m - 1$ by doubling modulo $2m - 1$, fixes the final position, and has the multiplicative order of 2 as return time (the two-card identity is a trivial edge case). More cards can therefore have the same or even a smaller return time.

A useful additional contrast

Random shuffling asks about a distribution over possible orders. This perfect shuffle is one precisely defined permutation repeated deterministically. Its return time says nothing by itself about how mixed a random shuffle is. Do not introduce chance unless it becomes a separate, explicitly changed rule.

Mathematical lineage and reuse

Optional adult bridge: coprime divisibility

If positive a, b are coprime and $a \mid bt$, compare base strips a, b with scaled strips at, bt . Repeatedly subtract the smaller base strip from the larger, doing the same to the scaled strips. Both scaled lengths stay multiples of a . The positive base sum decreases, so the process ends with equal lengths c . Common divisors are unchanged: a divisor of x, y also divides $x-y, y$, and conversely by adding. Thus $c=1$, since the original lengths were coprime. The scaled process ends at t , which is therefore a multiple of a . Conversely, $a \mid t$ plainly implies $a \mid bt$.

Try 5,3 first: $5-3=2$, then $3-2=1$. On scaled strips this leaves $3t - (5t - 3t) = t$. Equivalently, $2 \cdot 3t - 5t = t$; if $5 \mid 3t$, the remainder is a multiple of 5. Ask the child to explain each removal. This is additional readiness, not assumed knowledge of prime factors.

For $k > 0$, set $d = \gcd(n, k)$, $n = da$, $k = db$. Return means $a \mid bt$; the lemma gives $a \mid t$, hence first return n/d . Every start has this length, so there are d loops. For $k=0$, every point is fixed. If only the 3-and-2 case was explained, record that proof and label the general rule conjectured or given as a theorem. Full strip prompts and the Week 9 application: `plans/coprime-divisibility-scaffold.md`.

Sources and boundaries

Judson, *Abstract Algebra: Theory and Applications*, Ch. 4, §4.1, Theorem 4.13, pp. 48–49, gives the cyclic order formula: textbook. Our rings are its additive version. **Diaconis, Graham, Kantor (1983)**, *The mathematics of perfect shuffles*, §2, Lemma 1, p. 177, proves the out-shuffle order via doubling: original paper. The extra develops this small return-time case and position rule, not the paper's shuffle-group classification. The plan has full research references. Rozhkovskaya, Lesson 2, problems 2.3–2.7, supplies the shift-cipher entry. *Math Circle by the Bay*, preface viii–x, supports manipulatives, changing pace, and reserves; Rozhkovskaya Lesson 8 records copying slowing the activity. Our rings, proof scaffolds, and hour are adaptations. Lowell Handout 6, problem 6.7, and Handout 7, problem 7.8, use a different card-ordering solitaire. Ask about recognition; do not assume this shuffle was taught.

Use history and checks

Record actual steps, ring sizes, and extensions used as F04-K/M/U/X-v3. Keep 26-letter Caesar messages as a later alternative, not extra arithmetic for early finishers. Future years can compare prime/composite ring sizes, affine maps, in-shuffles, or perfect-shuffle groups; avoid replaying the same small cases. `verify.py` checks all 12 shifts, the 10-letter routes, message examples, the 18-ring predictions, shuffle rows, position formulas, and first return times for 6, 8, 16 cards. The facilitator proofs establish why the patterns hold.

v3 checks and observations

Added ring experiments

K 1–2: from circle land triangle,square,star,circle; from triangle land square,star,circle,triangle. **3–4:** circle and square share the two-hop orbit; triangle and star share the other. Changing from circle to square never reaches triangle. **5–7:** the three-hop, hidden shift, and inverse results are on page 2. Read and scribe the shape records; writing is not a gate.

Middle 1: +2 takes A to C and I to A. **2:** +3 takes B,I,J to E,B,C. **3–4:** messages and incompatible clue are on page 2. **5–7:** C follows C,E,G,I,A,C, the A-route; D follows D,F,H,J,B,D, the B-route. The two markings are preserved by every +2 turn. **8:** full +3 route is on page 2. **9:** +5 gives A,F,A and B,G,B, length 2. **10:** +2 then +8 gives A,C,A; B,D,B; I,A,I; J,B,J. Each starting letter returns despite the shorter orbits.

Upper: new comparisons

1: +4 sends 0,9,2,8 to 4,1,6,0. **4:** +8 traverses each +4 loop in reverse. **5:** +3 gives three loops of length 4, +6 six loops of length 2, +5 one loop of length 12. **7:** full table is on page 2. **8:** eighteen-place +6 route is 0,6,12,0; +5 is 0,5,10,15,2,7,12,17,4,9,14,1,6,11,16,3,8,13,0. **9:** distances at turns 1–4 are 4,8,12,16 and 8,16,24,32; first whole laps are turn 3 in both. **10:** +4 then +8 returns all starts, with intermediate positions 4,5,9,3. **11:** both orders total +11, sending 0,2,9 to 11,1,8. **12:** in general shifts +a,+b compose as +(a+b), wrapping modulo 12.

Extra: position scaffolding

1–6: page 3 gives all eight-card rows, map, loops, doubling proof, and fixed last position. Position traces from starts 1,3,7 are 1,2,4,1; 3,6,5,3; and always 7. **7:** six-card rows are 012345,031425,043215,024135,012345. **8:** sixteen-card loops are (0),(15),(1,2,4,8),(3,6,12,9),(5,10),(7,14,13,11). The full return is 4. **9:** 6 and 16 cards both take 4; 8 cards take only 3. Ask for a proof after these checks: position 1 rules out earlier returns, and the same remainder-1 multiplier fixes all other positions below the last.

Evidence for the next revision

Record actual step sizes/decks attempted, mistaken starting-point counts, skipped-versus-landed confusion, success using a route record, and card-label/position confusion. Note exact adult rescue and what children wanted more time for. Distinguish observations from hypotheses, and cases checked from arguments proved. The extra and general theorem may remain unused; this is not incomplete work. v3 is unpiloted.

A city is a mathematical object

Question: How much information forces one arrangement? Children progress from a complete list to a uniqueness proof to a lower bound on clues. These are questions about permutations, Latin squares, and determining sets.

Preparation and prerequisites

Use snap cubes: six per youngest child, eighteen per middle child, forty for the upper child: **112 cubes**, plus a few for the launch. Prebuild towers if time is short. Numeral slips can record a second solution after the physical attempt. Start with page 1 for each child (seven student sheets). Keep continuation masters and this guide ready; copy the next page when the group needs it. Build on the tabletop or quick scrap-paper grids with one-inch cells. The smaller printed grids record solutions; they are not sized to hold every brand of cube.

K-1: compare heights/count to 3; no independent reading; recognize different orders. Middle: count to 3 and follow row/column rules; adult may read; distinguish an example from a complete list. Upper: count to 4; track constraints, cases, and uniqueness; no formal algebra. Extra: understand a 4-by-4 Latin square, coordinate labels, and a pigeonhole argument.

Launch

Let towers be handled first. Show 2,1,3 from the left. Ask which towers a tiny person sees; touch each visible tower. Walk to the other end. Only after this introduce the separate city rule: every row and column uses each height once. A clue counts visible towers, not the first tower's height or the number of cubes.

A shared hour with seven children and two adults

0-10: explore materials freely. 10-15: brief launch and rules. 15-35: main investigation. 35-40: movement/reset; preserve work. 40-55: continue or choose an optional proof. 55-60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

K-1 stop: Build all six skylines and describe their views. **Optional proof:** organize by first tower to show completeness; explain the impossible one-at-each-end view.

Middle stop: Build both top-clue cities and select one with the second clue. **Optional proof:** certify the list, then swap unseen rows/columns to rule out every one-clue puzzle.

Upper stop: Explain the complete five-clue case table. **Optional proof:** check all five deletion certificates and prove the three-clue alternative forces the same city. The computer's minimum claim is separate.

Complete lists and minimum clues

K-1: six skylines, not six unrelated puzzles

The complete list (left count, right count) is: 123 (3,1), 132 (2,2), 213 (2,1), 231 (2,2), 312 (1,2), 321 (1,3). Fix the first tower: each of three choices leaves two orders. This proves completeness without factorial notation. One visible tower from both ends is impossible with distinct heights: the tallest would have to occupy both ends. Both 132 and 231 show two from each end.

The two 2-by-2 cities are 12/21 and 21/12. The top-row left clue 2 chooses the former. If row/column rules are too much, keep investigating rows; a complete skyline list is already substantial mathematics.

Middle: exactly two cities, then exactly one

Top clue 3 forces first column 1,2,3. Top row is either 123 or 132. Column exclusions force the respective completions:

1	2	3
2	3	1
3	1	2

1	3	2
2	1	3
3	2	1

The right clue 2 on row 2 keeps the first. Deleting that clue leaves both. Deleting only the top clue permits 123/231/312 and 312/231/123 (swap rows 1 and 3).

A genuine minimum proof: No single visibility clue can determine an entire 3-by-3 Latin square. If it sees a row, swap the two other rows; if it sees a column, swap the two other columns. The clue stays true and the square changes. Two clues suffice in our example, so the minimum over all uniquely solvable 3-by-3 visibility puzzles is two.

Hints and evidence

K: fix the first tower and exchange the other two. Middle: point down the first column, then try both top rows. Ask what rules out another answer. Accept a complete physical case split; long written prose is unnecessary. If children only solve the city, ask for one clue deletion and an alternative, saving the minimum proof for conversation.

A human certificate for five clues

Upper: the unique solution

Both the opening and the five-clue puzzle have rows **1234/2341/3412/4123**. Here is a small exhaustive argument for five clues. Bottom row starts with 4 and has right visibility 2, so 3 must be last. Only 4123 and 4213 remain. Keep both: the bottom clues alone do not rule either out; row 3 and row 2 do that next.

Row 3 has two views of 2. Its six possible rows are **1423, 2143, 2413, 3142, 3241, 3412**. Cross out candidates conflicting with the bottom row. The surviving pairs and possible row 2s are:

Bottom row	Row 3	Row 2 with right view 2
4123	3241	none
4123	3412	2341
4213	3142	none

Check the final column by listing row permutations avoiding the two entries already in each column, then testing the right view. The top row is forced by missing column heights. This proves uniqueness without trusting a solver.

Deleting clues: witnesses

Deleting L3, L4, R2, R3, R4 gives **2, 2, 6, 12, 20** solutions respectively. L3 means the left clue on row 3. One alternative per deletion proves necessity:

Delete	Alternative city, top row first
L3	1234 / 3412 / 2341 / 4123
L4	1234 / 4123 / 3412 / 2341
R2	1324 / 2431 / 3142 / 4213
R3	1324 / 3142 / 2431 / 4213
R4	1234 / 2143 / 3412 / 4321

On optional upper page 4, check each certificate is Latin, keeps all four retained clues, and differs from the target. Restoring the clue before the next deletion keeps the five arguments separate. The five-clue set is **minimal**, but it is **not minimum**: left clues 4,3,2 on rows 1,2,3 uniquely determine this same target. First row is 1234. The rows with left view 3 are 1243,1324,1342,2134,2314,2341; all but 2341 repeat a column entry. With rows 1 and 2 fixed, the possible row 3s are 3412 and 4123; only 3412 has left view 2. The last row is forced.

The script exhaustively verifies that three is the minimum *for this target city among subsets of its 16 perimeter clues*. That computation is not a proof that every 4-by-4 city needs three. For children, the concrete three-clue alternative already disproves the claim that minimal means minimum.

Trades, bounds, and research

Extra: a lower bound with an escape move

Each bold block has form ab/ba . Exchanging a, b inside it preserves the multiset in each affected row and column. With at most three entry clues, one of four disjoint blocks has no clue. Switch it: a distinct Latin square keeps every given entry. Thus at least four *entry* clues are necessary for this square.

Hitting these four blocks is not sufficient. Clues at $(1,1), (1,3), (3,1), (3,3)$ leave rows 2 and 4 unclued; swapping those rows keeps all four clues. Any uniquely determining set must meet *every* trade. Conversely, the changed cells in any alternative completion form a trade; hitting every such trade rules out alternatives.

Keep clue types separate: Previous pages use outside visibility clues; the extra uses fixed entries. Bounds do not transfer automatically.

Where the mathematics goes

Undergraduate combinatorics studies permutations, Latin squares, constraint satisfaction, and hitting sets. Joy Morris's *Combinatorics*, chapter on Latin squares develops Latin-square structure; our visibility questions add a different constraint system.

Hatami and Qian, *Teaching dimension, VC dimension and critical sets for Latin squares* (2016), §1 and Theorem 1.2, study sets of *entries* determining a square and prove a quadratic lower bound for sufficiently large orders. Our trade argument is a small instance of the question, not their asymptotic theorem. The paper states a sharper conjecture; no current open-status claim is needed here.

Teaching sources and returning children

JRMF *Skyscrapers Teaching Guide*, PDF pp. 1–4: physical viewpoints, city rules, and reasoning; pp. 13–14: blank boards. Our clue instances and investigations extend that format. *Math Circle by the Bay*, printed pp. ix–x: manipulatives, extra challenges, and themes at different depths. Lowell Handout 2.1–2.2 seating constraints are related prior material, not evidence children have solved these cities.

Use F05-K-v3/M-v3/U-v3/X-v3. Prepared, not recorded as taught. Save trade classification, Latin transversals, and larger skyline recurrences for later years. Rebuild with `build.sh`; `plans/verify-week-05.py` checks the finite claims independently of the explanations.

Using the revised sequence

Classroom evidence and this new proposal

Revision v3, September 27, 2026: **unpiloted**. Week 1 feedback reported too little work before new ideas, unclear next actions, and simultaneous adult demand. It did not report a Week 5 trial. These revisions apply that guidance by giving each viewpoint several attempts and each later representation a job. Preserve children's actual records; do not equate an unfinished packet with a failed session.

Launch and handoff

After handling the cubes, gather everyone around towers 2,1,3. Touch the towers visible from each end; record 2,1,3 left to right and write its two view counts. Then change to 1,3,2 and let a child check. Distribute first pages only after children can make and record that action. Demonstrate the row/column city rule locally when a group reaches a grid; the K group can spend the whole session on rows.

K pages 1–2 provide repeated views and a complete list. Problem 1 has views (2,1), Problem 2 has (2,2). Problem 3 accepts 123 for three; 312 or 321 for one; 132,213,231 for two. Problem 6 has no solution; preserve two unsuccessful attempts before discussing the tallest-at-both-ends argument. Page 3's city rule is a separate reserve.

Middle page 1's four rows have view pairs (3,1), (2,1), (2,2), (1,3). Any Latin city and any whole-row swap meet Problem 2. Pages 2–3 stage the top clue, a second clue, and a deletion, with separate grids. Stop after the two top-clue cities if comparison still needs time. Ask the completeness and unseen-row/column-swap proofs orally only after the children have built the alternatives.

Upper page 1's row views are (2,2), (2,2), (1,2), (1,2). Page 2 isolates possible rows before page 3 organizes them into cases. The complete lists and table on this guide's page 3 answer Problems 4–7. Pages 4–5 are optional clue investigations. The covered-clue views for Problem 9 are respectively **3,3,3,3,4**; each violates the restored target clue while retaining the other four. A smaller construction and a lower bound are different claims. Preserve the minimal/minimum discussion in conversation after the checked deletion records and three-clue construction.

Extra and practical pacing

Extra page 1 tests four individual switches before entry clues appear on page 2. For any three circled cells, at least one of four blocks is untouched; any such switch works. For Problem 4, use the original city and **1234/4321/3412/2143**, obtained by swapping rows 2 and 4. Each has the four fixed entries. Invite the four-block lower-bound and every-trade arguments from page 4 of this guide after these concrete checks.

Allow 15–35 minutes for first attempts, a 35–40 movement break, and 40–55 for the current idea or one selected continuation. Keep 55–60 for sharing. Do not start two adult-dependent proof discussions together. Offer pointing, building, or dictation for every recording task.

Record after use

Record exact problem numbers attempted, viewpoint/row-rule errors, any instructions needing repeated adult rescue, what children chose to continue, and explanations actually given. Keep observations separate from hypotheses and proposed revisions. Source support: *Math Circle by the Bay*, pp. ix–x; Rozhkovskaya, Lessons 3,7,8, “At the lesson.” These support flexible pacing and representation after attempts; the exact sequence is our adaptation.

Useful questions; provably few tests

Question: Can we design a method that works for every secret, and prove no shorter method can always work? This is decision-tree complexity with exact-position feedback.

Prepare and check the rules

Use two kinds of counters (mark R/B or distinct symbols), paper position strips, and three folded-paper screens. Use 12 counters of each kind for K and middle, and 16 of each kind for upper so six candidate rows and a test can remain visible. Keep all records. A parent may write dictated guesses and scores. First score visible secret RBR against RRR (2), BRB (0), and RBR (3). Repeated colors are allowed. Give only the **total correct positions**; no wrong-place-color feedback.

Start with page 1 for each child (seven student sheets). Keep the remaining pages as continuation masters, including any earlier records children will compare. Print the extra only when selected. K: no reading; match positions and count 0–2; eliminate impossible examples. Middle: counts to 3, optional R/B reading; compare one changed feature and complete cases. Upper: counts to 4; worst-case reasoning, symmetries, and branching. Extra: 0/1 notation and small equations; variables can be replaced by counters.

Count the same thing throughout

A **test** is a scored query. Once exactly one secret remains, naming it is free in this investigation. Original game rules require actually submitting the secret, potentially one extra turn. Our exact optima for 2, 3, 4 positions are **2, 3, 4 tests to identify**, not claims of those same turn limits for winning submission. Stop early if a score already identifies the secret.

A shared hour with seven children and two adults

0–10: explore materials freely. 10–15: brief launch and rules. 15–35: main investigation. 35–40: movement/reset; preserve work. 40–55: continue or choose an optional proof. 55–60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

K-1 stop: Build a complete two-test method. **Optional proof:** for any first test, build two secrets scoring one, so one test cannot always work.

Middle stop: Explain why the three-test method works. **Optional proof:** combine the one-B triple obstruction with color renaming to cover every first test. Without renaming, claim only the RRR-first obstruction.

Upper stop: Explain why four tests always suffice. **Optional proof:** defend the triple and four-candidate branches separately, then use symmetry to prove optimality. Choose one older group's proof conversation at a time.

Two and three places: complete proofs

K-1: elimination becomes a decision tree

RR scores 0 only for BB, 2 only for RR, and 1 for RB or BR. In the last branch test RB: scores 2 or 0 distinguish them. So two tests suffice. One cannot always work: for every first test, two secrets match in exactly one position. Build those counterexamples.

Middle: a three-test upper bound

Let baseline RRR score s , the number of Rs in the secret. Flipping one test position R to B changes its contribution from 1 to 0 if the secret there is R, or 0 to 1 if it is B. The new score is $s - 1$ or $s + 1$, never unchanged. Tests BRR and RBR reveal positions 1 and 2. Their number of Rs subtracted from s determines position 3. This procedure covers all eight secrets.

Middle: why two tests cannot guarantee success

After RRR scores 2, candidates are BRR, RBR, RRB. The complete second-test table is:

Test	BRR	RBR	RRB
RRR	2	2	2
RRB	1	1	3
RBR	1	3	1
RBB	0	2	2
BRR	3	1	1
BRB	2	0	2
BBR	2	2	0
BBB	1	1	1

Every row repeats a score; some answer leaves two candidates. Conceptually, the three secrets differ only in the position of one B. A test only reveals whether that B falls inside its B positions, giving at most two outcomes. For the numbered symmetry step, physically turn BRB into RRR by swapping R/B at places 1 and 3 in both secrets and queries. An equal pair stays equal and an unequal pair stays unequal. Every score is preserved. Do this at the B positions of any first query; it becomes RRR, so the lower bound covers all first queries and all adaptive methods.

Hints

Make candidate rows tangible. The table is optional once a child explains the one-B structure and two possible scores; further entries add no proof. Test a troublesome pair first. Ask what score a test would earn for each candidate. A test need not itself be a possible secret to reveal useful information.

Four tests: upper and lower bounds

Upper bound

RRRR and three separate single-position flips determine places 1–3 by score changes; the baseline R count determines place 4. Four tests suffice. Generally this uses n tests to identify an n -bit secret, at most $n + 1$ turns if winning submission is required.

A difficult answer that stays consistent

By independent color renaming, take the first test to be RRRR. Answer 2, leaving six two-B secrets. Classify the second test by its B count. Position permutations and complementing a test (replacing score s by $4 - s$) reduce all cases:

0 or 4 Bs: no information; six remain.

1 B (BRRR): scores 3 or 1 each leave three candidates. Take score 3: BBRR, BRBR, BRRB. Ask: *Which position is fixed? What is the only feature that changes?* They share the first B; the other B occupies one of three places. Any last test has a fixed first-position contribution and only two possible scores on the remaining one-B triple. It cannot separate all three. The score-1 branch has the same structure; 3-B tests are complementary.

2 Bs (BBRR): score 4 leaves BBRR, score 0 leaves RRBB, score 2 leaves BRBR, BRRB, RBBR, RBRB. Choose the four-candidate branch. For any last test with k Bs and a two-B secret, let j positions have B in both. The score is $2 - k + 2j$. There are at most three possible scores, so four candidates cannot all be separated.

Every second test has a consistent branch that needs more than one further test. Thus three tests cannot guarantee identification; four is optimal. The lower bound uses actual score structure, not merely five possible answers per test.

Facilitating a hard proof

Let the child physically sort six code cards into score piles. Ask them to make the largest pile small, then play the stubborn setter. Upper pages 1–2 supply a satisfying stop. Pages 3–4 are an optional adult-supported lower-bound investigation; the final-query triple argument is different from the four-candidate counting argument. If full optimality is too much, obtain a proof of the four-test method and one failed three-test plan. Record that boundary honestly.

Relationship to Week 3: Here the hidden object is an ordered binary string, queries may adapt, and the goal is a worst-case measurement optimum. Score structure and adversarial lower bounds deepen earlier identification ideas.

Five bits and a research landmark

Extra: decode and prove

The example is **10110**, with scores 2,2,2,0. Let w count secret 1s and s_0, s_1, s_2, s_3 be scores in printed order. Then $w = 5 - s_0$, and

$$x_4 + x_5 = \frac{s_1 - s_0 + 2}{2}, \quad x_3 + x_5 = \frac{s_2 - s_0 + 2}{2}, \quad x_2 + x_5 = \frac{s_3 - s_0 + 2}{2}.$$

Each 0-to-1 change gains one match for a secret 1 and loses one for 0. If any pair-sum is 0 or 2, it fixes x_5 , then x_2, x_3, x_4 ; w fixes x_1 . Otherwise all three sums are 1. If $x_5 = 0$, the middle bits are 111 and w is 3 or 4. If $x_5 = 1$, they are 000 and w is 1 or 2. The ranges do not overlap. So w fixes the case and x_1 .

This proves four *preselected* tests distinguish all 32 secrets. The page claims sufficiency, not an independently proved five-place optimum. Matching scores encode binary-hypercube distances; distinguishing landmarks form a resolving set.

Undergraduate and research connections

Decision trees, adversary arguments, Hamming distance, and information bounds belong to discrete mathematics and algorithm analysis. The small optima here have explicit upper and lower proofs.

Cáceres et al., *On the Metric Dimension of Cartesian Products of Graphs* (2005 preprint; 2007 publication), §2, PDF p. 3, gives exactly our four five-bit landmarks. §6, PDF p. 8, equates exact-position static Mastermind with resolving sets in Hamming graphs. This directly adapts a finite construction from a research paper, with our elementary decoding proof.

Knuth, *The Computer as Master Mind* (1976–77), gives a five-guess strategy for commercial four-place, six-color Mastermind; see Knuth's author page and strategy link. That game also has wrong-place-color feedback. Its result does not directly apply here. The shared idea is minimizing the worst surviving candidate set.

Teaching lineage and future use

JRMF *Crack the Code*, PDF pp. 2–4 supplies binary exact-position play and the $n + 1$ -turn challenge. The optimum/adversary tasks and extra deepen it. *Math Circle by the Bay*, pp. ix–x supports manipulatives, varied pace, and optional challenges.

Use F06-K-v3/M-v3/U-v3/X-v3; prepared, not taught. Save more colors, noise, and average-case analysis for later. `plans/verify-week-06.py` searches all adaptive query trees for small optima and checks all 32 extra signatures.

Make the scores usable first

Revision v3: unpiloted

The September 27 revision applies the Week 1 classroom guidance; no Week 6 classroom outcomes have been reported. We added repeated visible scoring, contrasting single-position changes, physical candidates, and separate recording space before trees and adversary arguments. These are design hypotheses to test, not claims of observed success.

A concrete whole-group launch

After handling and copying counters, keep RBR visible. Build RRR alongside it and point to matches in positions 1 and 3. Say only “2” and record “RRR: 2.” Compare BRB (0) and RBR (3). Let a child check a test and dictate the log entry. Hide a secret only when this action is understood. Give first pages to groups; demonstrate branches locally after candidate filtering, never as the opening notation.

K page 1: against RB, tests RR,RB,BR,BB score **1,2,0,1**. Page 2: RR gives **2,1,1,0** against RR,RB,BR,BB. On remaining RB/BR, RR scores (1,1), RB scores (2,0), and BR scores (0,2). Page 3 introduces the tree to reuse those trials. It gives one test for RR/BB and two for RB/BR. The one-test lower bound remains an oral follow-up after testing the tree on all four secrets.

Middle page 1: visible RBR has scores 2,0,3. Page 2’s rows BRR,RBR,RRB,BBB have score triples **(2,3,1), (2,1,3), (2,1,1), (0,1,1)**. Ask what a gain/loss says about the changed position before requesting the general method. A working three-test method is a satisfying stopping point. Page 3 uses a small selection of tests plus children’s own before the optional full-table argument. In Problem 6, swapping positions 1 and 3 sends RRB,BBB,BRR to BRR,RBR,RRB; every old and new score is 2. Checking examples motivates the symmetry proof on page 2 of this guide; examples alone do not prove that all first tests are covered.

Upper page 1: score pairs for RBRB,BRBR,RRRR,BBBB are **(2,1), (2,3), (4,3), (0,1)**. Upper page 2’s four-test records for RBRB,BRBR,RBBB,RRRR are **(2,1,3,1), (2,3,1,3), (1,0,2,2), (4,3,3,3)**. Keep each comparison anchored to RRRR. Naming is free throughout.

Plan the adult conversations

Use 0–10 handling, 10–15 shared launch, 15–35 play/comparison, 35–40 movement, 40–55 current work or one continuation, 55–60 sharing. Starting-page print plan: seven sheets plus continuation masters. K may stop after consistent filtering; middle/upper may stop with a reliable method and its explanation. Return earlier score records alongside later pages. Do not require every table row if the child can already explain its structure.

Observe whether children score positions consistently, compare with the correct baseline, and use records to choose a next test. Record actual examples and claims established, not merely pages distributed. Distinguish observed difficulties, hypotheses about them, and proposed changes. The Week 1 review and Rozhkovskaya’s Lessons 3,7,8 motivate this order; the exact tasks are our adaptations.

New tables and proof handoffs

Upper pages 3–4: checked score records

In candidate order BBRR,BRBR,BRRB,RBBR,RBRB,RRBB, second-test rows are:

BBBB	3	3	3	1	1	1
BRRR	4	2	2	2	2	0
BRRB	3	3	1	3	1	1
BBBR	2	2	2	2	2	2

For the triple BBRR,BRBR,BRRB in Problem 7, final tests RRRR,RBRR,RBBR give **(2,2,2), (3,1,1), (2,2,0)**. For the quartet BRBR,BRRB,RBBR,RBRB in Problem 8, RRRR,BRRR,BBRR give **(2,2,2,2), (3,3,1,1), (2,2,2,2)**. Children's added tests vary; check by counting matches directly.

After these trials, offer the complete lower-bound conversation on page 3 of this guide. A finite collection of failed final tests is not enough. Establish separately why the moving-B triple allows at most two scores and why a two-B quartet allows at most three. Then cover every second test by B-count/position symmetry, and every first test by independent color renaming. Without the last step, report only an RRRR-first obstruction. Keep this distinction in the use log.

Extra: three pages before the general decoding argument

Page 1 keeps all four probes visible through three examples and two actual games. Probe order is 00000, 00011, 00101, 01001. The provided secrets have records:

00000	5	3	3	3
11111	0	2	2	2
10010	3	3	1	1

Page 2 first isolates two changed positions. Secret pairs 00,01,10,11 have 00/11 match counts (2,0),(1,1),(1,1),(0,2); changes are **-2,0,0,+2**. Thus score change reveals pair sums 0,1,1,2. The decoding example has total 1-count 3 and pair counts **1,1,0**; it gives 10110 as before. Keep physical counter pairs available before writing equations.

Page 3 gives the ambiguous pair-count case its own examples. If all three pair sums are 1, the four words are **01110,11110,00001,10001**, with total 1-counts **3,4,1,2**. Their four-score records are respectively (2,2,2,2), (1,1,1,1), (4,4,4,4), (3,3,3,3). Any one score distinguishes these four, though these are not all 32 possible secrets. Revisit the hidden games using the counts; then offer the general two-case proof from page 4 of this guide.

Keep the research conclusion precise: four fixed probes suffice to distinguish all 32 five-bit words; this packet does not prove a five-place optimum. The hypercube/landmark translation belongs in the discussion after decoding works. Binary distance is number of mismatches, hence five minus score.

Games are recursive mathematics

Question: Which player has a strategy against every reply? Can a finite chart justify an infinite pattern? What information is needed when two games are combined?

Prepare and launch

Bring 25 counters per group, large move cards **1 or 2**, **1, 3, 4**, and **1, 3, 5**, and pencils. Start with page 1 for each child (seven student sheets). Keep the remaining pages as continuation masters, including any earlier records children will compare. Print the extra only when selected. Pre-numbered charts reduce copying. Two distinct mats identify the extra's two piles. For the memory reserve, bring a paper strip and a movable four-cell frame (or two folded scraps).

Let children move and count counters before rules. Demonstrate 1-or-2 from three counters; let a child move and check legality. Remove an allowed positive number no larger than the pile. Taking the last counter wins. Older children then receive the clearly different 1,3,4 card. Do not announce the losing-number pattern at the launch.

K: count/remove 1 or 2 and alternate turns; no reading; examine both replies. Middle: counts and subtraction to 14 (physical allowed); adult can read; distinguish one winning move from all losing options. Upper: subtraction/remainders to 28, with groups of seven as a concrete alternative; two-direction argument and induction. Extra: small nonnegative labels, smallest-missing operation, and decreasing two-pile games; no binary arithmetic needed for the printed proof.

A shared hour with seven children and two adults

0–10: explore materials freely. 10–15: brief launch and rules. 15–35: main investigation. 35–40: movement/reset; preserve work. 40–55: continue or choose an optional proof. 55–60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

K-1 stop: Demonstrate both replies that make three a trap. **Optional proof:** group counters in threes and explain the strategy for all multiples of three.

Middle stop: Defend W/L labels through seven. **Optional proof:** refute greedy play at eight and justify both winning moves at ten.

Upper stop: Conjecture losing remainders 0,2 and defend sample labels. **Optional proof:** show both remainder obligations and termination; reserve the paper-window argument for a later step.

Endgames build the whole strategy

K-1: groups of three

From 3, go second: answer 1 with 2 or 2 with 1. From 6, do the same, leaving 3; repeat. From 7 take 1 to leave 6; from 8 take 2; from 9 go second. Multiples of 3 are losing for the next player. Each pair of turns removes one group of three; the second player takes the last. Two demonstrated replies are a proof without a formula.

Middle: backward reasoning

For moves 1,3,4, positions 0–20 are:

Pile	0	1	2	3	4	5	6
W/L	L	W	L	W	W	W	W
Pile	7	8	9	10	11	12	13
W/L	L	W	L	W	W	W	W
Pile	14	15	16	17	18	19	20
W/L	L	W	L	W	W	W	W

Zero is L for the player who would move next. If a position has a move to L, take it: the position is W. If all moves go to W, each gives the opponent a winning strategy: it is L. Every move decreases the pile, so this bottoms out. It is backward induction, not a tally of who happened to win.

From 5 take 3 to leave 2; from 6 take 4. From 7 all moves leave 6,4,3, which are W. Taking 3 from 8 leaves 5, and the opponent can take 3 to leave 2. Take 1 instead, leaving 7. From 10, taking 1 (leave 9) or 3 (leave 7) wins. Both are winning choices; taking 4 leaves W position 6.

Hints that preserve investigation

Ask about one or two counters. For a disputed label, lay out all legal next piles. Check whose turn the label describes. If a child calls 7 winning because they won once, replay as the opponent with correct replies; return the discovery to the child.

Returns: Lowell Handout 2.3 and Handout 3.1–3.3 used 1-through-3 and 1-through-5. Those establish previous materials, not mastery. Handout 3.1 mistakenly calls 5 losing after observing it can move to losing 4; with those old rules, 5 is winning. Use the checked definitions here.

An infinite proof from finite structure

Upper: losing remainders 0 and 2 modulo 7

Proposed L piles are $7k$ and $7k + 2$, including zero. Subtraction of 1,3,4 never connects these remainders. Every other remainder has a legal move into the set:

Remainder	1	3	4	5	6
Take	1	1	4	3	4
New remainder	0	2	0	2	2

Each move is legal even for the smallest positive representative. These are both proof obligations. Starting outside the set, move into it; every opposing reply exits; move back. The decreasing pile ends at zero, reached by your move. Starting inside gives the second player this strategy. The argument covers all piles, not just checked examples.

Change to 1,3,5

Every move is odd and changes parity. No move connects even piles. Every positive odd pile can take 1 to reach even. Even piles are L and odd piles W: second wins from 6 or 100, first from 7. The proof structure persists while the invariant changes. This is different from taking any number 1 through 5.

Why finite subtraction games eventually repeat

Copy labels onto a straight strip and frame four consecutive entries. To add the next label, inspect entries one, three, and four places back: the rightmost, second-from-left, and leftmost cells. Slide the frame one cell right after adding it. Match two windows and physically extend both to see the same next label. For 1,3,4, once four labels exist, the next depends only on those four. There are at most $2^4 = 16$ W/L windows. Among 17 windows some pair agrees. Identical windows give identical next labels; induction forces all future labels to repeat. This proves *eventual* periodicity, not repetition from zero or period seven. The remainder proof supplies those stronger facts here.

For any finite move set with maximum m , use m -label windows, starting after enough entries exist to use the full recurrence. Grundy values (next page) are bounded by the number of allowed moves, so their windows also eventually repeat. Infinitely many allowed moves do not provide a fixed finite-memory argument.

Adult attention and optional depth

The oldest child should defend a disputed label and a strategy to an adult. If the remainder investigation takes the whole session, save the finite-window argument and extra. Record which proof actually appeared; a completed chart alone is not understanding why a pattern continues.

Beyond W and L: combining games

Extra: smallest-missing labels

For 1,3,4, labels for piles 0–10 are **0,1,0,1,2,3,2,0,1,0,1**. Thus (1,1) loses; (1,4) wins by taking 1 from the second pile to leave (1,3). Both (1,3) and (4,6) lose; (5,6) wins by taking 1 from the first pile to leave (4,6). W/L status alone forgets information needed when games combine.

The two-pile proof: By the smallest-missing definition, no option from label a has label a , and every smaller nonnegative label occurs among its options. A move from equal labels changes exactly one, destroying equality. With unequal labels, change the larger to the smaller; the definition guarantees such a move. Both losing-position obligations hold. Total size decreases, and terminal (0,0) has equal labels. This is the two-component case of Sprague–Grundy theory. For three or more piles the rule is bitwise XOR.

Undergraduate and research lineage

Ferguson’s UCLA Math 167 course includes take-away, Nim, graph games, and sums. His *Game Theory*, Part I, §1.3–1.4 develops winning/losing and subtraction games; §3.2 defines Sprague–Grundy values and §4.2 treats sums. Our proofs are concrete instances.

Shenxing Zhang, *On the linearity of the periods of subtraction games* (2024), §1, Lemma 1.1 and Conjectures 1.2–1.3, investigates changes in periods and preperiods when one move is added to a fixed set; Theorem 1.4 proves the predicted behavior in several families. Our seven-period example is elementary; predicting whole families is research. We describe the paper’s conjectures, not an exhaustive survey of their current status.

Sources, reserves, and verification

JRMF *Countdown Teaching Guide*, PDF pp. 1–2: play-first questions; pp. 4–5: W/L reasoning; p. 8, challenge 4 explicitly uses 1,3,4. Our parity comparison, finite-window scaffold, and extra extend it. *Math Circle by the Bay*, printed pp. ix–x supports manipulatives and local extensions; the two-adult arrangement is ours.

Use F07-K-v3/M-v3/U-v3/X-v3, prepared, not taught. Save misère play, other move sets, and three-pile XOR for later. `plans/verify-week-07.py` checks 1,001 single-pile values and independently solves two-pile games with each pile at most 30; proofs above establish the unbounded claims.

Stages and new concrete checks

Revision v3: unpiloted

No Week 7 session evidence is supplied. This revision applies the Week 1 feedback by putting repeated play and legal-option records before W/L, grouped counters before remainders, actual window extensions before a finite-state proof, and two-pile games before number labels. These design choices await classroom evidence.

After handling the counters, gather everyone for 1-or-2 play from three. Let a child make one legal move, say the number taken, and record the number left. Check the final move. Then distribute first pages. Introduce older groups' 1,3,4 move card explicitly. A win in one game is an observation, not a guarantee against every reply. Demonstrate W/L only when a child has met that distinction in small endgames.

K and middle

K page 1 records actual games. Page 2's replies are 2 after 1 and 1 after 2, from either 3 or 6; from 6 they leave 3. Page 3: choose first from 7 (take 1), first from 8 (take 2), second from 9. From 12, paired-turn replies leave **9,6,3,0**. The general groups-of-three argument remains an oral follow-up, not a required written conclusion.

Middle page 1's pile 2 can only leave 1; pile 4 can leave 3,1,0. Page 2 builds the labels through 7 with complete legal destinations. Stop there if all-versus-one reasoning still needs play. Page 3: from 8 the options are 7L,5W,4W; from 10 they are 9L,7L,6W. Therefore 8 has just the winning take-1 move; 10 has winning takes 1 and 3. For self-chosen strategy cards through 14, use the checked chart on page 2 of this guide.

Upper: each representation has a task

Page 2 Problem 4: 14 reaches 13W,11W,10W; 16 reaches 15W,13W,12W; 15 reaches 14L,12W,11W; 17 reaches 16L,14L,13W. Page 3 builds these as two sevens plus leftovers 0,2,1,3. The remainder table has L at 0,2. From 0, subtraction of 1,3,4 reaches remainders 6,4,3; from 2 it reaches 1,6,5. W remainders have the winning moves on this guide's page 3. From 24 take 1 to leave 23, a losing pile; later replies maintain remainder 0 or 2. Discuss both losing-position obligations and legal boundary cases after the table, preserving the full proof.

Page 4's changed game has even losing piles. In the old game's windows, starts 0 and 7 read LWLW; next is W and the slide gives WLWW. Starts 1 and 8 read WLWW; next is W and the slide gives LWWW. Other valid matching windows differ by seven in the displayed chart. Only after these actual extensions introduce 16 possible windows and the eventual-periodicity argument. It does not by itself prove period seven or repetition from zero.

Extra: use the labels after meeting the problem

Page 1 outcomes: (1,1) L, (1,3) L, (1,4) W, (4,4) L. From (1,4), the winning first moves take 1 or 3 from the pile of 4 to leave (1,3) or (1,1). Page 2's g-values are on this guide's page 4. From pile 4, next labels are 1,1,0 (missing 2); from 5 they are 2,0,1 (missing 3). Page 3 adds (3,4), won by taking 1 or 3 from the pile of 4 to equalize labels.

For (4,6), first moves from pile 4 of 1,3,4 can be answered from pile 6 by 3,3,4 respectively. First moves from pile 6 of 1,3,4 can be answered by: take 1 again from that pile; take 1 from pile 4; take 4 from pile 4. Each restores equal labels. Then offer the general mex/equality proof; do not infer it solely from these examples.

Start with seven first-page sheets and continuation masters. Keep earlier charts with later pages; stop after a defended small strategy, one general argument, or a tested label comparison. Record actual moves, misunderstood rules, adult rescue, voluntary continuations, and claims justified, separating observations from explanations. Flexible pace and action before tables are supported by *Math Circle by the Bay*, pp. ix–x, and Rozhkovskaya, Lessons 3,7,8; this sequence is our adaptation.

From play to a useful record

F08 v3, September 2026: prepared and unpiloted. These changes apply the organizer's Week 1 feedback; there is no reported Week 8 classroom outcome. The goal is a strategy discovered through repeated concrete games. Pages are a menu for more than one possible meeting.

Prepare a small starting set

Print page 1 for each child at their starting level: three K–1, three middle, one upper (7 sheets). Keep one master of each continuation and the 3-page extra; copy only what is needed. Middle page 3 reuses its earlier board. Bring about 60 counters, five plates or marked pile areas, pencils, and scraps labeled 1,2,4,8. Printed pile mats are usable counter areas; reset and reuse them.

K–1: adult reads; compare distances/count to 3. Middle: count to 7 and track moves, with adult scribing if needed. Upper: small subtraction and odd/even; bundles introduce binary notation. Extra: organize lists and compare differences; use a calculator only for the final supplied formula. Choose by prerequisites, not grade.

Common launch before separate starts

0–10: let everyone move counters and build piles. 10–15: gather at one visible rook board. Show a one-square right move, a longer down move, and the same token reaching the star. Invite a child to make a legal move; have another mark its start and finish. Say: “One token, right or down, at least one square; reaching the star wins.” Show an illegal diagonal only if needed. Do not reveal equal distances yet. Place the token at (2,1) and build matching piles of 2 and 1. Slide right one square while removing one counter from the 2-pile. State that pile games remove any positive number from just one pile; taking the last counter wins. Then distribute starting tasks. This one correspondence shows both actions without revealing a strategy.

15–35: play the contrasting starts. 35–40: movement/reset. 40–55: repeat, compare, or introduce the next record only when it answers the children's question. 55–60: share one move with its reason and tidy. The timetable is an adaptable proposal, not a source-prescribed duration.

Parent stays mainly with K–1. Organizer alternates middle/upper about every five minutes, leaving a specific next game. Triplets rotate two opponents and a checker. The upper child needs actual games and discussion with an adult.

Pacing, hints, and stopping

Ask for a legal move before a strategy. Then ask what is left to the star, or compare an equal and an unequal start. K can stop after showing the two center moves. Middle can stop with an argued board pattern or the board/pile correspondence. Upper can stop after the six replies to (1,2,3), or after balancing concrete bundle records. No general proof is a printed completion requirement; pages 2–3 retain the proofs for conversation. The extra can stop after its small map.

Checked two-pile reasoning

K-1 and middle board answers

Distances (a, b) mean squares remaining right and down. A rook move reduces exactly one positive coordinate. The star is $(0, 0)$: the previous player has already won. On the 3×3 board, the center $(1, 1)$ is losing. Both possible moves allow an immediate reply to the star. Top-left $(2, 2)$ is also losing; copy the amount removed in the other direction.

On the 4×4 board, the traps form the top-left to bottom-right diagonal: $(3, 3), (2, 2), (1, 1), (0, 0)$. Every unequal pair is winning: reduce the larger coordinate to the smaller. The middle start $(3, 2)$ has the unique winning move to $(2, 2)$.

A complete proof, with counters

If two piles have the same size, every move makes them unequal. If they differ, reducing the larger to the smaller is legal and restores equality. Thus a player who hands over equal piles can always restore equality after an opponent's move. The total number of counters strictly decreases, so play must finish. The player restoring equality takes the final counters; the opponent receives $(0, 0)$ and cannot move. These two properties establish all traps, on rectangles as well as squares.

Any unequal two-pile start has exactly one winning first move (specifying the pile and amount). Matching piles have none. Children can use $(5, 2)$ as their designed puzzle: remove 3 from the 5-pile.

Why three piles change the problem

$(1, 1, 1)$ is winning: remove one pile and leave $(0, 1, 1)$. Even total is insufficient: $(1, 1, 2)$ has total 4 but is winning by removing the whole 2-pile. $(1, 2, 3)$ is losing. Here are all first moves and one reply; pile order is kept fixed.

After move	Reply leaves	After move	Reply leaves
$(0, 2, 3)$	$(0, 2, 2)$	$(1, 2, 0)$	$(1, 1, 0)$
$(1, 0, 3)$	$(1, 0, 1)$	$(1, 2, 1)$	$(1, 0, 1)$
$(1, 1, 3)$	$(1, 1, 0)$	$(1, 2, 2)$	$(0, 2, 2)$

Each reply leaves two equal piles and one empty pile. This gives a physical proof before binary notation. The middle group can stop here or join the upper bundle experiment.

The binary invariant

Upper answers

Binary records: $1 = 001$, $2 = 010$, $3 = 011$, $4 = 100$, $5 = 101$, $6 = 110$, $7 = 111$. $(1, 2, 3)$ and $(2, 4, 6)$ have even column sums. $(1, 2, 4)$ does not; reduce 4 to 3. For the table: $(2, 4, 6)$ is balanced; $(3, 4, 5)$ can become $(1, 4, 5)$; $(1, 5, 7)$ can become $(1, 5, 4)$; $(3, 5, 6)$ is balanced. Other correct balancing moves should be accepted.

For $(7, 10, 12)$, the Nim sum is 1; reduce 7 to 6. For chosen first piles a, b , the unique third pile is $a \oplus b$: each binary digit is forced by whether the first two digits agree.

Why the test is sufficient, for any number of piles

A move changes exactly one integer. At least one of its binary digits changes, so at least one previously even column becomes odd. No legal move can connect two balanced positions.

If the Nim sum s is nonzero, choose its highest 1-bit, say the 2^k column. Some pile x has a 1 there. Replace it by $y = x \oplus s$. Higher bits stay fixed, this bit becomes 0, and lower bits may change. The sum of all smaller bundles is $2^k - 1$, so $y < x$: the move is legal. Every odd column is toggled once, every even column is unchanged, and the result is balanced. The decreasing total ensures termination. These two facts prove exactly the same win/lose structure as in the two-pile proof.

Do not physically permit removing an arbitrary bundle while leaving a larger number of counters: the game is still ordinary removal from one pile. The bundle display is a way to calculate a legal smaller result.

Extra: Wythoff, checked results and proof boundary

The old equal-pile trap fails because both piles can be emptied in one diagonal move. For $0 \leq a \leq b \leq 7$, the traps are $(0, 0)$, $(1, 2)$, $(3, 5)$, $(4, 7)$. Greedy pairs through $n = 5$ are $(0, 0)$, $(1, 2)$, $(3, 5)$, $(4, 7)$, $(6, 10)$, $(8, 13)$.

Distinct pairs have disjoint coordinates and distinct absolute differences, so neither a single-pile move nor an equal-take move connects them, including reversed pile order. This proves only the no-move-between-traps condition. To prove completeness one must also show every outside position reaches a listed pair. Sample moves: $(2, 4) \rightarrow (2, 1)$, $(5, 8) \rightarrow (4, 7)$, $(8, 12) \rightarrow (6, 10)$. The golden-ratio formula is a cited theorem, not a conclusion from five data points.

Where the mathematics leads

Undergraduate direction and genuine research

Two piles give a strategy-preserving change of representation from a grid to an impartial game. Binary column parity is addition in a vector space over the two-element field. The proof has a useful algorithmic form: recognize the losing set, prove no move stays inside it, and construct a move into it from everywhere else. Formal algebra vocabulary can wait.

Charles Bouton's historical paper, *Nim, a Game with a Complete Mathematical Theory*, *Annals of Mathematics* 3 (1901–02), pp. 35–39, is the original research source for binary analysis. Our normal-play proof is written out on page 3, so the circle does not depend on reading the historical notation. Caltech-hosted paper.

Fraenkel and Peled, *Harnessing the unwieldy MEX function*, *Games of No Chance 4*, MSRI Publications 63 (2015), pp. 77–94: §1, pp. 77–78 states the Wythoff rule, greedy pairs, and golden-ratio formula; §5 develops efficient computation for generalized complementary sequences. Author research chapter.

The extra packet meets an actual research theme: a simple rule change can replace bitwise arithmetic with complementary sequences and a question of efficient computation. It does not reproduce the paper's generalized algorithm or claim an open problem has been solved by the classroom experiment.

Local sources and teaching choices

JRMF, *Rook's Move Activity Guide*, PDF pp. 3–6: legal moves, exploration before advice, losing positions, and the Wythoff connection. We replace the old upper king variant with three-pile Nim; the king version remains an untaught reserve unless the use log says otherwise.

Math Circle by the Bay, preface printed pp. viii–x (PDF pp. 9–11), supports common themes with adjustable depth, manipulatives, explanations, and extra challenges. Rozhkovskaya, Lesson 7 “At the lesson,” EPUB part0017.xhtml, highlights checking that children understand legal moves. The seven-child schedule and bundle prompts are our adaptations.

Lowell Handout 2, problem 2.3, and Handout 3, problems 3.1–3.3, include subtraction-game history; Week 7 also studies constrained subtraction. Here an entire pile may be removed, and the important new step is multiple piles plus binary balance. Reuse is deliberate, not evidence that every child has seen these exact starts.

Record and verify

IDs F08-K/M/U/X-v3. Record the actual starts, rule, level, strategies explained, and whether Wythoff was attempted. Keep unused stages for later years. `verify.py` checks all three-pile states through size 12 and Wythoff states through 30 by independent backward recursion. Those finite checks support the examples; the proofs establish the unlimited claims.

Use each record for a reason

Core sequence and specific hint ladders

K pages 1–2: compare an immediate win with the center, then equal top-left distances with an unequal start. Demonstrate drawing a move from square center to square center. When stuck, enumerate the center's two legal moves with the token. Page 3 is a separate translation: build two equal piles, copy a removal, then change to unequal piles. Stop before this transition if rook play remains engaging.

Middle pages 1–2: play eight purposeful starts before the T/W map. Demonstrate the (right-distance, down-distance) record at (2,1) with the actual token. The first table records game outcomes, not proved winning types. Before mapping, classify (1,0) and (1,1) together: a W has one move to T, a T has every move to W. Ask children to challenge one proposed letter; a lost game alone proves neither T nor W. Page 3 physically matches two slides to removals before asking for equal-pile replies. Page 4 adds three piles only after equal-pile play is understood. The six preprinted outcomes prevent copying from consuming the investigation.

Upper pages 1–2: compare adjacent starts and test the even-total guess. Two equal piles plus an empty pile is useful, but (2,4,6) has no immediate move to that form: failing to find one is a reason to seek another representation. The six replies to (1,2,3) can be a complete stopping point before binary. Page 3 groups real counters before recording 0/1 columns. Model 5 as 4+1, then let the child build. If notation obscures the game, leave bundle cards visible beside ordinary piles. Page 4 tests the same record on four cases and actual opponent moves. Page 5 introduces the highest-odd-column recipe only after the repair problem is familiar.

Proof reserve: ask why a move from balance must alter a column; then why the highest changed bundle outweighs all lower bundles together; finally use decreasing total for termination. Preserve both directions of the losing-set argument. Naming XOR is optional and follows the child's record, not the launch.

Extra and observation record

Play equal and unequal starts before the classification grid. Work only far enough to reveal a second nontrivial trap; finish the grid later if useful. Demonstrate sorting after a move: (2,4) can become (2,1), recorded at (1,2). The greedy sequence is a compressed record of already encountered traps. Distinct coordinates/differences prove no internal moves; completeness is a separate requirement. Supply the golden-ratio result as a theorem only if reached.

After teaching, record exact starts/pages, who needed a rule repeated, useful or burdensome records, child conjectures, and requests to continue. Separate observations from hypotheses and adult-supplied theorems. Mark unused pages unpiloted; do not infer uptake from printing.

Additional v3 solution checks

Board and pile instances

K1 immediate-above-star is (0,1), winning immediately; the center is (1,1), losing. From top-left (2,2), right 1 is answered down 1; right 2 by down 2, and vice versa. The middle-left start (2,1) can move right 1 to (1,1). Equal pile starts (2,2),(3,3) use matching replies; (3,1) removes 2 from the larger pile.

Middle trial starts (1,0),(2,1),(3,1),(3,2),(2,3) are W; (1,1),(2,2),(3,3) are T. Unique moves to equal distances respectively leave (0,0),(1,1),(1,1),(2,2),(2,2). At (3,2), right 1 corresponds to piles (2,2); down 2 to (3,0). In pile Problem 6, (3,2) reduces to (2,2), (1,4) to (1,1); equal starts require an opponent's move first. The six replies in middle Problem 8 and upper Problem 4 are listed on page 2.

Upper bundle work

Problem 3: (1,1,1) removes any single 1; (1,1,2) removes the 2. Neither (1,2,3) nor (2,4,6) reaches two equal nonempty piles and an empty pile in one move, since no pair is initially equal. Their totals 6 and 12 are even, but the total-4 winning example (1,1,2) refutes even-total sufficiency.

Problem 5 records in 4/2/1 order are 1=001, 2=010, 3=011, 4=100, 5=101, 6=110, 7=111. Problem 6 column counts for (1,2,3) are (0,2,2); for (1,2,4), (1,1,1). Reduce 4 to 3. Problem 8 checked moves are on page 3; multiple balancing moves can occur.

Problem 10: (4,6,7) has odd 4 and 1 columns. Legal repairs are 4 to 1, 6 to 3, or 7 to 2. In (7,10,12), only the 1 column is odd; reduce 7 to 6. Problem 11 third piles are 7 for (2,5) and 5 for (3,6); each bit is forced. Accept a child-designed zero-XOR start with a demonstrated legal reply.

Extra instances

Problem 1: (2,2) and (2,3) are winning, moving respectively to (0,0) and (1,2); (1,2),(3,5) are traps. Problem 2's four outcomes are (0,2),(1,1),(1,0),(0,1); empty the remaining pile or take 1 from both. Problem 3 grid has exactly (0,0),(1,2),(3,5),(4,7) as T. Problem 4: (2,4) to (2,1); (4,6) to (3,5); (5,7) to (3,5), all legal. Greedy/formula rows and final large moves are on page 3. For each tested pair, single-pile removal would need a shared coordinate; equal removal would need the same difference. Neither occurs between distinct listed traps.

Bounces before mirrored rooms

F09 v3, September 2026: prepared and unpiloted. Applying Week 1 feedback is a design inference, not evidence these billiards pages have been taught. Tracing comes before unfolding; unfolding gets its own concrete work before arithmetic formulas.

Preparation and access

Print page 1 at each child's starting level (3 K–1, 3 middle, 1 upper: 7 sheets), with continuation masters and the 3-page extra available. Copy K page 3 separately when folding; cut between its two panels first. Keep earlier traced boards for later comparisons. Bring counters, pencils, rulers, scrap grid paper, and scissors for an adult. About 10 minutes' preparation; no actual ball.

K: adult reading/tracing allowed; child chooses turns. Middle: count lengths and multiples through 12, then count whole rooms. Upper: multiples, division and parity; gcd/lcm and the coprime lemma are optional adult-supported continuations. Extra: fractions and coordinates; supply irrationality if necessary. Oral reasoning and adult scribing are valid.

Common launch before separate starts

0–10: let children handle counters and trace grid diagonals freely. 10–15: gather at a visible 2×4 table. Move diagonally from the black dot to the right wall. Ask a child to show the next move that stays inside and still heads upward. Dot that bounce and circle the ending corner. Show that interior line crossings do nothing. Everyone now knows the physical action and what to record; distribute different starts only afterward. Save mirror rooms for a later question about predicting a path.

15–35: trace and compare. 35–40: movement/reset. 40–55: revisit a puzzling path, fold one copy, or continue the room-count investigation. 55–60: share one checked prediction and tidy. These times are adjustable. Parent mainly supports K; organizer alternates middle/upper visits and leaves a specific next rectangle. Rotate mover, tracer, checker, so each child acts.

Hints and useful stopping points

First check the bounce rule. Then enlarge or trace a difficult section while the child directs. Introduce mirrored rooms by physically folding a line across one wall, not by announcing the lcm. K may stop with contrasting corners. Middle may stop with matched bounce/crossing numbers, then later predict corners. Upper may stop with room-count parity and a verified bounce rule. General coprime division, no-bottom-left and complete inverse-design proofs remain adult conversations on pages 2–3. A returner can supply one familiar trace and proceed to the representation they have not used.

Checked paths and corner rule

K-1 and middle answers

For width w , height h : 2×2 ends top-right with no bounce. 2×4 follows $(0, 0)$, $(2, 2)$, $(0, 4)$, ending top-left; 4×2 follows $(0, 0)$, $(2, 2)$, $(4, 0)$, ending bottom-right. Folding the two width-2, height-4 rooms turns the straight diagonal through $(2, 2)$ into the first of these paths.

For 2×3 , the folded path is $(0, 0)$, $(2, 2)$, $(1, 3)$, $(0, 2)$, $(2, 0)$. In copied rooms, the first simultaneous wall meeting is $(6, 6)$. The four middle rows have $(L, \text{widths}, \text{heights}, \text{corner})$:

Table	L	Widths	Heights	Corner
2×3	6	3	2	bottom-right
3×6	6	2	1	top-left
3×9	9	3	1	top-right
4×6	12	3	2	bottom-right

A middle design reaching top-right after a bounce can be 2×6 (two bounces); top-left can be 2×4 . Accept any verified design.

Upper theorem with all hypotheses

Take positive integer w, h and a 45-degree launch. Reflections unfold to $y = x$. A corner occurs when the common coordinate is both a multiple of w and of h . Its first positive value is $L = \text{lcm}(w, h)$. First use room counts $m = L/w, n = L/h$. If they shared $d > 1$, distance L/d would already cross m/d widths and n/d heights: an earlier corner. Thus they are coprime; both cannot be even, ruling out bottom-left without a gcd formula.

Optional arithmetic bridge. Write $w = ga, h = gb$ with coprime a, b . Any common multiple is gat with $b \mid at$. To justify $b \mid t$, begin with $a = 2, b = 3$: both $3t$ and $2t$ divide into threes, so their difference t does too. In general, run repeated subtraction on strips of lengths a, b and, alongside them, at, bt . Subtract the shorter from the longer until equal. Common divisors are unchanged, so the coprime base strips end at 1 and the scaled strips at t . Both scaled lengths remain multiples of b throughout; hence $b \mid t$. The least positive t is therefore b , and $L = gab, m = b, n = a$. The shared plan scaffold supplies physical bundle examples. If this step is supplied, record it as a given theorem.

An odd number of whole widths folds to the right, even to the left. An odd number of whole heights folds to the top, even to the bottom. Thus odd a , odd b gives top-right; even a , odd b gives bottom-right; odd a , even b gives top-left. Both cannot be even, so first arrival at bottom-left is impossible. Scaling both dimensions preserves the reduced pair and corner.

Upper rows: 4×6 : $L = 12, m = 3, n = 2$, bottom-right; 6×9 : 18, 3, 2, bottom-right; 6×10 : 30, 5, 3, top-right; 8×14 : 56, 7, 4, bottom-right. Raw parity of the original lengths does not determine the endpoint.

Bounces, inverse design, and slopes

Count crossings, then design

Before the final corner the line crosses $m - 1$ vertical room walls and $n - 1$ horizontal ones. With the optional lemma these are $b - 1$ and $a - 1$. No crossing belongs to both lists: that would be an earlier corner. So the number of bounces is $a + b - 2$ (not counting launch or terminal corner). Path length, if wanted, is $\sqrt{2}L$; L itself is horizontal distance, not path length.

For top-right after exactly four bounces, a, b are coprime odd positives with $a + b = 6$. Candidates $(1, 5), (3, 3), (5, 1)$ reduce to just $(1, 5), (5, 1)$ because $(3, 3)$ is not coprime. Any positive integer scaling works, such as 2×10 . Top-right after three bounces is impossible: two odd numbers have even sum, so $a + b - 2$ is even. The latter is a parity obstruction, not a failed search.

Extra packet: rational and irrational directions

The extra uses a *unit square* and physical slope $s = p/q$ in lowest terms. A positive lattice point (u, v) on $y = (p/q)x$ satisfies $qv = pu$, so its first occurrence is (q, p) . By the coprime lemma on page 2, q divides u ; the first positive choice is $u = q$, giving $v = p$. The endpoint's horizontal side is right iff q is odd; its vertical side is top iff p is odd. There are $p + q - 2$ bounces. Slope $2/3$ ends bottom-right with 3 bounces; $3/2$ ends top-left with 3.

For slope $\sqrt{2}$, an integer corner (u, v) with $u > 0$ would give $\sqrt{2} = v/u$, impossible. If requested, prove irrationality: reduce a supposed fraction v/u first. From $v^2 = 2u^2$, v is even; substituting $v = 2k$ forces u even, contradicting the reduction.

The other launch $(0, 1/2)$ with slope 1 cycles through $(1/2, 1), (1, 1/2), (1/2, 0), (0, 1/2)$. It avoids corners but follows just a diamond. No-corner does not imply dense. Density of the $\sqrt{2}$ -slope trajectory launched from $(0, 0)$ is a separate theorem. Under our stopping rule, arbitrary starts need care: from $(0, 2 - \sqrt{2})$ the same slope reaches unfolded $(1, 2)$, folds to bottom-right, and stops.

Keep the conventions straight

A general rectangle with physical slope s becomes a square after rescaling axes, with normalized slope sw/h . Rationality of that normalized slope controls the square model. The extra fixes $w = h = 1$ to avoid silently mixing those slopes. "Rational polygon" in research means its angles are rational multiples of π ; it does not mean a rational launch slope. Our corner-start paths stop when they hit a corner, so do not call those stopped paths periodic billiard orbits.

A doorway into dynamics

Mathematical direction and primary sources

Unfolding replaces a piecewise reflected path by a straight line on a lattice. The corner problem is divisibility, coprime integers, and parity; the inverse problem asks for a complete classification of parameters. These are substantive undergraduate number-theoretic and geometric ideas in a tangible representation.

Masur and Tabachnikov, *Rational billiards and flat structures*, published in *Handbook of Dynamical Systems* 1A (2002), pp. 1015–1089. In the linked author manuscript, §1.3 gives unfolding, §1.4 develops the unit-square/torus model, and §1.5 extends the construction to flat surfaces. Later §4 concerns periodic orbits. Author manuscript.

The classroom genuinely performs the first construction used in the research: reflect copies, then study straight motion. It does not establish the deeper recurrence or ergodicity theorems. The extra explicitly separates avoiding corners from density; the latter is supplied specifically for the origin-start $\sqrt{2}$ path under our stopping convention.

Downloaded sources and documented prior use

JRMF *Billiards Geometry*, PDF p. 2, asks for corner, bounce-count, and path-length rules. Lowell Handout 10, problems 10.2–10.7, includes 7×5 , 4×3 , 8×10 , scaling, consecutive sides, $1 \times n$, and $2 \times \text{odd}$ families. Handout 11, problems 11.4–11.5, includes height-five rectangles and plus-shaped tables. Do not describe small rectangle families or scaling as wholly new for returning children.

Our changed emphasis is a proof through physical unfolding, an exact first-corner criterion, a bounce formula, and inverse existence/impossibility design. This deliberately promotes an old reserve into the core. Plus-shaped tables remain a later branch; do not claim the rectangle formula covers them.

Math Circle by the Bay, preface printed pp. viii–x, supports manipulatives and revisiting ideas at different depths. Rozhkovskaya, Lesson 3 “At the lesson,” EPUB part0013.xhtml, supports attempts and child explanations before formal displays; Lesson 8’s account warns about copying pace. Preprinted grids and optional adult tracing are our response, not a procedure claimed to be in those sources.

Keep a useful history

IDs F09-K/M/U/X-v3. Record actual dimensions, launch slopes, whether unfolding was used, which formulas were explained, and which claims stayed conjectural. A prepared extension is not a taught activity. Future variants can change slopes, starting points, or polygons instead of enlarging the same arithmetic table.

`verify.py` independently traces all 900 rectangles with sides 1–30, compares the corner and bounce formulas, and checks the inverse-design list. The general proofs are on pages 2–3; finite checks alone do not establish them.

Representation gates and records

Move, mark, compare; then unfold

K pages 1–2: the square has no bounce, tall and wide tables turn at different wall types, and the height-6 table adds a second bounce. The child chooses directions even when an adult draws. Page 3 offers two separate folds, one across a vertical wall and one across a horizontal wall. Cut between the two diagrams; fold the right half left for Problem 5 and the top half down for Problem 6. A spare sheet or tracing paper makes the reflected segment visible. Folding is a follow-up to the paths, not an opening explanation.

Middle: page 1 gives three contrasting traces. Page 2 numbers bounces and unfolded crossings so the new picture has a concrete matching job. Demonstrate the first crossing (2,2); let the child match the rest. Page 3 gives separate 3-by-6 and 3-by-9 diagrams before translating parity into an endpoint table. Have the child actually move back and forth along a room edge as counts rise. Page 4 introduces lists of multiples to predict a larger table without drawing all copied rooms, then supplies design grids with stated bounds. No freehand rectangle is required to be metrically exact if the counted grid is correct.

Upper: page 1 contrasts a square and swapped side lengths. Page 2 gives original and copied 4-by-6 rooms side by side; the child matches actual bounces with crossings. Page 3 uses a scaled copy and two even-sided tables with different endpoints, preventing raw parity from being mistaken for the rule. The false distance 24 forces attention to the first corner. Page 4 counts separate wall lists before a formula. Page 5 compares three candidate designs, including the reducible 3-by-3 case. The two genuine four-bounce designs therefore contrast with an early-stop failure.

Preserve the proof as a conversation

The least common distance argument, “earlier corner if both room counts share a divisor,” and wall-count proof can be shown directly on records. Only then offer reduced sides and the coprime-divisibility scaffold. If the child uses the formula but cannot justify its lemma, record it as supplied. The complete inverse-design proof and irrationality proof remain on pages 2–3. No child needs all these arguments in one hour.

The extra gives five rational trials, including $2/4$ vs $1/2$, before the general p/q claim. The midpoint diamond gives a concrete counterexample to “no corner implies dense.” Keep the origin hypothesis on the supplied irrational-slope density theorem. Arbitrary offsets and general rectangles need the qualifications on page 3.

After the meeting

Record actual tables/slopes, bounce-rule misunderstandings, whether folding clarified the path, which records were useful, and questions children wanted to pursue. Distinguish an observed difficulty from a proposed explanation; record conjecture, explained proof, and supplied theorem separately. These v3 changes remain unpiloted until actual use is reported.

Additional v3 solution checks

K and middle

K Problem 3: 4-by-2 ends bottom-right after 1 bounce; 2-by-6 follows $(0,0),(2,2),(0,4),(2,6)$, ending top-right after 2. Problem 5 folds the straight 4-by-4 diagonal into the 2-by-4 path; Problem 6 folds it into the 4-by-2 path.

Middle page 1 corners/bounces: 2-by-2 top-right/0; 2-by-4 top-left/1; 4-by-2 bottom-right/1. Problem 4 unfolds 2-by-3 crossings at distances 2,3,4, ending at 6. The corresponding folded bounces are $(2,2),(1,3),(0,2)$. Whole-room counts are 3 widths and 2 heights. Problem 6: 3-by-6 reaches $(6,6)$, counts $(2,1)$, top-left, 1 bounce; 3-by-9 reaches $(9,9)$, counts $(3,1)$, top-right, 2 bounces. Endpoint rows alternate right/left and top/bottom.

Problem 8: 4-by-6 has $L=12$, counts $(3,2)$, bottom-right; 2-by-6 has $L=6$, counts $(3,1)$, top-right; 3-by-3 has $L=3$, counts $(1,1)$, top-right. Design examples within the 6-by-6 grids: 2-by-6 top-right after 2 bounces, 2-by-4 top-left after 1. A square fails the “after at least one bounce” condition.

Upper

Problem 1: 2-by-3 ends bottom-right/3 bounces; 3-by-2 top-left/3; 3-by-3 top-right/0. Problem 4 for 4-by-6 has vertical distances 4,8,12 and horizontal 6,12: first common 12. The 6-by-9 scaling factor is $3/2$; L scales 12 to 18 while counts and corner stay the same. 6-by-10 ends top-right; raw even side lengths do not determine corner. Problem 8's distance 24 predicts bottom-left, but the path stops at 12 first.

Problem 9: 2-by-3 vertical list 2,4, horizontal 3, total 3; square both empty, total 0; 4-by-6 lists 4,8 and 6, total 3; 6-by-10 lists 6,12,18,24 and 10,20, total 6. The count is $(m-1)+(n-1)$. Doubling dimensions doubles all distances, preserving counts. Problem 12: 1-by-5 and 5-by-1 have four bounces and end top-right; 3-by-3 stops immediately. Problem 13 may use 2-by-10. For Problem 14, three bounces would give $m+n=5$; top-right requires both m,n odd, impossible. Page 3 gives the full reduced-pair classification.

Extra

Slopes $1/2$, $2/3$, $3/2$ first reach $(2,1),(3,2),(2,3)$, with corners top-left, bottom-right, top-left and bounce counts 1,3,3. Slope $2/4$ equals $1/2$ and stops at $(2,1)$, not $(4,2)$. Slope $3/4$ first reaches $(4,3)$, top-left, 5 bounces. Problem 4 examples: $1/5$ or 5 for top-right/4; $2/3$ for bottom-right/3. For the midpoint diamond, $(1/2,1/2)$ and a small neighborhood of it are never visited. Absence of corner hits alone does not establish density; the separate stated theorem does.

Walk before counting degrees

F10 v3, September 2026: prepared and unpiloted. Week 1 feedback motivates more contrasting walks and records before parity, splicing or optimization. No Week 10 classroom outcome has been supplied. Maps are connected and undirected; parallel bridges are allowed, self-loops are not used.

Preparation and access

Print page 1 for each child at the appropriate starting level (3 K–1, 3 middle, 1 upper: 7 sheets). Keep continuation masters and the 3-page extra; copy on demand. Retain earlier maps for comparisons. Bring seven moving counters, about 35 used-edge markers, pencils and scraps for route strips. Optional strings and paper islands enlarge a map; no calibrated printing is needed.

K: adult reads; distinguish an island from a bridge and track used bridges. Middle: counts to 4 and odd/even; adults may write route lists. Upper: follow route lists, pair arrivals/departures and compare constructions with lower bounds. Extra: add small lengths, compare three choices; shortest connections are a new idea introduced through actual walks.

Common launch before separate starts

0–10: children freely build little maps or move along printed ones. 10–15: gather at a triangle. Walk a counter A–B–C–A, putting a marker on each used bridge. Say: “Visit every bridge once; revisiting an island is allowed.” Show how the route list records each visited island. Invite a child to try another start and mark a used bridge. Add a branch and start an attempt, then distribute starting pages. Save odd/even language for after contrasting attempts.

15–35: try walks and compare starts. 35–40: movement/reset. 40–55: repair a map, splice a second loop, or improve a delivery route at the children’s current stage. 55–60: share a route and a reason; tidy. Parent mainly stays with K; organizer alternates middle/upper about every five minutes. Triplets rotate walker, marker, and checker. These times are flexible proposals.

Hints, gates and stopping

First ask which bridge is unused and where the counter is. Next compare a successful and a stuck attempt. Only then pair arrivals/departures. K may stop with two-leaf versus three-leaf attempts and one repair. Middle may stop with forced house endpoints or a repaired map. Upper may stop with two concrete splices; delivery becomes a new stage with a changed rule (repetition allowed). The full Euler construction and postman lower bounds are discretionary adult conversations on pages 2–3. Children need not finish every page to encounter substantial mathematics.

One-walk maps: checks and proof

K-1 answers

A triangle has a closed route, e.g. $A-B-C-A$. Add a branch $A-D$: a trail is $D-A-B-C-A$, with endpoints D and A . At a degree-1 leaf, a walk cannot arrive and then leave by a different unused bridge. Thus the three-leaf star cannot have a trail covering every edge once: it would need three endpoints, while a walk has at most two.

Adding a bridge $B-C$ between two leaves repairs the star: $D-A-B-C-A$ is a complete trail. It begins/ends at the remaining leaf D and center A . Accept another pair of leaves with relabeled route.

Middle and upper map answers

House degrees: $A=2, B=2, C=3, D=3, E=2$. One trail is $D-A-B-C-D-E-C$. Every complete trail must start at one of C, D and end at the other. No closed Euler trail exists.

For the triangle with center D , all four vertices A, B, C, D have degree 3. It has no every-edge-once trail. Add a second AB bridge; now A, B are even and C, D odd. A valid trail is $C-A-B-D-A-B-C-D$, using the two distinct AB bridges separately. Middle children may draw the added bridge as a curve; a crossing without a dot is not a new junction.

Euler criterion with a construction

In any completed trail, arrivals and departures at an internal vertex pair. A closed Euler trail therefore has only even degrees; an open Euler trail has exactly two odd vertices, its endpoints. The necessity argument includes repeated visits to a vertex: each visit contributes another pair.

First physically splice the two triangles ABC and ADE sharing A . Mark AB, BC, CA used after $A-B-C-A$; AD, DE, EA remain. Insert the strip $A-D-E-A$ at a visit to A to obtain, for example, $A-D-E-A-B-C-A$. Trace it and mark all six roads. This one construction motivates the next general argument.

For sufficiency, assume a finite connected graph with at least one edge and all degrees even. Follow unused edges until stuck. At a vertex other than the start, arriving uses one edge of an even available count and leaves at least one to depart; so only the start can be where the walk stops. If unused edges remain, connectedness ensures that some vertex of the current closed walk touches unused edges (take a path toward an unused edge). Start a new closed walk there in the remaining even-degree graph and splice it into the first. Repeat. Finitely many edges guarantee completion.

For exactly two odd vertices, add one temporary edge between them, construct the closed Euler trail, then remove that temporary edge to obtain an open Euler trail. Parallel edges are allowed. This proves existence; it does not promise that every arbitrary greedy walk uses all edges without splicing.

Delivery routes and optimality

Upper: six unit-length roads

The triangle-and-center network is K_4 , with six unit roads. Every closed covering walk must repeat at least two road crossings: all four original vertices are odd, and each added traversal changes parity at its two endpoints only. A closed walk has even traversal-degree everywhere. Thus six original plus at least two repeats gives a lower bound of 8.

A matching route is A–B–C–D–A–C–D–B–A. It repeats AB and CD; all other roads occur once. The three possible odd-vertex pairings are AB+CD, AC+BD, AD+BC. Each adds two roads and creates an even connected multigraph, so each admits a closed Euler trail of length 8.

If finishing anywhere is allowed, at most two odd vertices may remain. At least one repeated crossing is necessary, giving a lower bound of 7. A–B–C–A–D–C–D–B attains 7, repeating CD.

For the design brief, a star with five leaves has six odd vertices: five degree-1 leaves and the degree-5 center. But its leaves are not adjacent, so it cannot meet the three-extra-step target. A workable design is two triangles joined by three corresponding cross edges (a triangular prism graph). All six degrees are 3. Duplicate those three cross edges; now all degrees are 4, so a closed tour exists with exactly three extra crossings, meeting the parity lower bound. Label the triangles ABC and DEF, with cross edges AD, BE, CF. One 12-step closed route is A–B–C–A–D–E–B–E–F–C–F–D–A. Children may find other designs.

Extra: six roads of different lengths

Original total is $2 + 3 + 7 + 1 + 6 + 1 = 20$. Distances between odd vertices are $d(A, B) = 2$, $d(A, C) = 2$ via D, $d(A, D) = 1$, $d(B, C) = 3$, $d(B, D) = 3$ via A, $d(C, D) = 1$.

Pair costs: AB+CD=3; AC+BD=5; AD+BC=4. Choosing the cheapest available AD first is not guaranteed optimal. Duplicate AB and CD to obtain minimum total 23. The same A–B–C–D–A–C–D–B–A route has length 23. Long roads AC and BD must still be serviced once; shortest connections are used only for the *extra* travel.

For a general lower bound, subtract one required copy of every road from a closed covering walk. The remaining multigraph has odd degree exactly at the original odd vertices. Its edges can be decomposed into paths pairing those odd vertices and closed loops: follow unused edges from an odd vertex to another odd one, remove that path, and continue, then remove cycles. Each path costs at least the shortest distance of its endpoint pair, and loops cannot improve a nonnegative-cost total. Thus every solution pays at least the best pairing cost. Duplicating shortest paths for that pairing attains the bound by Euler's construction.

The mathematics behind the workshop

Undergraduate graph theory

The optional proof conversation completes the Euler trail criterion, including its constructive direction; the physical splice is a satisfying earlier step. It distinguishes feasibility from optimization and separates an example route (an upper bound) from an obstruction (a lower bound). The upper activity turns parity into an algorithm for a small route-inspection problem.

Cornell CS 2800, Spring 2017, *Lecture 36: Paths*, gives the even-degree theorem and a cycle-splicing proof. Official course notes. Our maps, graded questions, and counter-based launch are original adaptations.

A real research connection

Edmonds and Johnson, *Matching, Euler tours and the Chinese postman*, *Mathematical Programming* 5 (1973), pp. 88–124, studies the weighted route-inspection problem using matching theory. The introduction describes the problem and its structure; §4 concerns a matching algorithm and §5 Euler tours. IBM publication record; Paper hosted by University of Michigan.

The extra packet performs an exact four-odd-vertex instance of the same reduction. It does not teach the general blossom algorithm, polyhedral proof, or directed/mixed variants. For $2k$ odd vertices, exhaustive pairings number $(2k-1)(2k-3)\cdots 1$; efficient weighted matching makes large instances practical. The research is a historical solved result, not a claimed present-day open problem.

Local sources and returners

Rozhkovskaya, *Math Circles for Elementary School Students*, M.4.1 “Make your own problem,” EPUB part0031.xhtml, inspires making under constraints. Its animal/figure/number prompt is different; our connected-map/parity specifications are our design. Lesson 3 “At the lesson” supports child attempts and explanations before an organizing display.

Math Circle by the Bay, preface printed pp. viii–x, supports adjustable depth and extra challenges. Our timetable and two-adult allocation are proposals for this roster.

Lowell Handout 10’s password-window puzzle is related to directed Euler/de Bruijn constructions; do not call graph traversal wholly new. Here the representation is an undirected bridge map, and the claims concern odd-degree obstructions and minimum repeated travel. Week 1’s flip graph concerned states and distance; here edges are roads that must all be serviced. Generic old workshop starters remain optional future reserves.

Use history and checks

IDs F10-K/M/U/X-v3. Record maps, repairs, exact routes, whether optimality was justified, and child-designed problems with their solutions. `verify.py` independently searches states (current vertex, set of roads already visited) and confirms closed/open/weighted optima 8, 7, 23. It also checks printed routes. The mathematical certificates above explain why the results hold.

Stages that can stop independently

Concrete attempts before the invariant

K: triangle versus branch, then two-leaf path versus three-leaf star, give contrasting successful and stuck walks. Demonstrate resetting used-edge markers before another start. The two clean star diagrams make repairs genuinely different trials. Ask about leaving a leaf only after children have met one. Their own small map is discretionary; no degree vocabulary is necessary for the endpoint argument.

Middle: page 1 gives routes and several house starts before page 2 requests degree counts. A failed route is recorded as an attempt, not a proof. At an internal visit, put two fingers on arrival/departure lines and repeat if the route returns to that dot. Page 3 gives two distinct repairs (AB and AC), with fresh maps and space for routes. Then join the remaining odd dots to make a closed version. Page 4 designs on supplied dots; preserve the child's solution as well as the map. If numbering roads is easier than a letter list, use it.

Upper: actual attempts precede parity. Page 2 splices a loop at A and then at an internal B visit, so insertion is not just a slogan or an abstract algorithm. Page 3 explicitly changes from every road once to every road at least once, allowing repeats. Try two routes before counting the extra copies. Page 4 makes three concrete pair repairs, then changes the endpoint requirement. Page 5 compares two six-odd maps: the count alone gives the same lower bound, but one attains it and one does not.

Proof reserves and extra

For Euler necessity, repeated visits still pair arrivals and departures; only endpoints can have one unpaired incident road. The two splices motivate, but do not prove, all-even sufficiency. Page 2 preserves the connectedness, finite-edge, leftover-even and temporary-edge arguments. For delivery, match a concrete route with a parity lower bound. Do not equate "six odd dots" with "three repeats suffice." The star is the counterexample.

The extra first totals actual weighted routes, then distinguishes a required long road from a cheap extra connection. Walk A–D–C to compare with AC; B–A–D to compare with BD. All original roads still require service. Only after that comparison introduce the three pairings. AD and CD tie for cheapest first pair but lead to different totals: greedy tie choice is not guaranteed safe. The augmentation decomposition proof on page 3 is an optional adult discussion after the child draws their own extras. Preserve shortest-path costs, not direct-road-only pairing costs.

Record actual learning

Save route lists and failed attempts, which rule needed rescue, whether paired fingers or strips helped, and which challenges children wanted to continue. Distinguish observations from hypotheses, construction from optimality proof, and explained results from supplied theorems. Record each actually attempted v3 page; the rest remain unpiloted. The exact cases/sequence are our Week 1-informed adaptation, not claims about prior students' behavior.

Additional v3 solution checks

K and middle

K triangle: A-B-C-A or B-C-A-B. Branch: D-A-B-C-A; B cannot be the endpoint of a complete route because it has even degree while A,D are odd. The two-edge path A-B-C works; the three-leaf star cannot. With BC added, use D-A-B-C-A. With BD added, use C-A-B-D-A. Each ends at the center and unjoined leaf. Child maps need direct checking, not one assumed answer.

Middle house attempts from A cannot finish every edge once; C and D work as opposite endpoints. The branch degrees are A=3,B=2,C=2,D=1. K4 degrees all equal 3. House degrees and route are on page 2. With duplicate AB, a trail is C-A-B-D-A-B-C-D. With duplicate AC, a trail is B-A-C-B-D-A-C-D. Add CD in the first repair, or BD in the second, to make all degrees even. Closed routes appear below. For the five-dot design, the existing house is one valid answer; adding CD makes every degree even. Many designs work.

Upper construction and costs

Problem 3: A-D-E-A-B-C-A. Problem 4: A-D-E-A-B-F-G-B-C-A. The degree at B changes from 2 to 4; all 9 roads appear once. Problem 5 optimum 8 and lower bound are on page 3. For that route, AB and CD are crossed twice; others once; every crossing-degree is 4. One extra copy fixes two of four odd degrees, so cannot make a closed walk possible.

Problem 8 closed routes: AB+CD uses A-B-C-D-A-C-D-B-A; AC+BD uses A-C-B-D-A-B-D-C-A; AD+BC uses A-D-B-C-A-B-C-D-A. Each costs 8. Problem 9 can duplicate CD, using A-B-C-A-D-C-D-B, length 7 with endpoints A,B. Problem 10 can add AD,BE,CF between triangles ABC and DEF and repeat those same three links; the 12-step route on page 3 uses every road. Problem 11's five-leaf star requires all five roads twice: A-B-A-C-A-D-A-E-A-F-A, length 10, five repeats. Each leaf requires both arrival and departure on its only road.

Extra route certificates

Problem 1: any complete route is admissible; optimal total is 23. Problem 2 original sum 20. Cheapest connections and all pairing totals are on page 3. Problem 5's best AB+CD route is A-B-C-D-A-C-D-B-A, total 23. AD+BC can use A-D-B-C-A-B-C-D-A, total 24. Choosing AD first forces BC, adding $1+3=4$; choosing CD first leaves AB, adding $1+2=3$. This refutes an unconditional cheapest-first guarantee, not every greedy run. Problem 7 for the optimum leaves separate AB and CD copies; their four endpoints are the original odd vertices. General augmentation pairing, including multi-road paths and nonnegative-cost cycles, is proved on page 3.